

Highlights

Determinants of Hyperparameter Invariance in Connectome Reservoir Computing

Miles Walter Churchland, Raul de Palma Aristides, Miguel Soriano, Anna Ritz, Greg Anderson

- The *C. elegans* connectome occupies a low hyperparameter-sensitivity regime as a reservoir.
- Higher task-agnostic capacity often coincides with higher sensitivity to reservoir hyperparameters.
- Raw spectral radius before normalization mediates much of this tradeoff across sign and shuffle controls.

Determinants of Hyperparameter Invariance in Connectome Reservoir Computing

Miles Walter Churchland^a, Raul de Palma Aristides^b, Miguel Soriano^c, Anna Ritz^d and Greg Anderson^e

^aDepartment of Computer Science, Reed College, Portland, OR, United States

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Abstract

Biological neural circuits maintain functional computation despite substantial variation in synaptic strengths, membrane properties, and inputs. Reservoir computing provides an artificial framework for studying this robustness: input signals are projected into a high-dimensional space by a fixed nonlinear dynamical system, called the reservoir. Unlike biological circuits, these systems can be highly sensitive to hyperparameters such as spectral radius, input scaling, leak rate, and bias. This paper asks which reservoir-network features support invariance to those parameter changes. Using the *C. elegans* connectome as a biologically grounded reservoir, we measure computational capability with four task-agnostic metrics—memory capacity (MC), information processing capacity (IPC), kernel rank (KR), and generalization rank (GR)—and define hyperparameter invariance as the coefficient of variation of these metrics across sweeps of spectral radius, input scaling, leak rate, and neuron bias.

To isolate contributors to invariance, we construct controlled perturbations that separately modify connectivity topology, weight-magnitude distributions, magnitude placement, and excitatory/inhibitory sign structure while preserving the remaining properties. Across these experiments, the empirical connectome consistently occupies a relatively low-variance regime.

The central result is a performance–invariance tradeoff: perturbations that raise task-agnostic computational capacity tend to make those capacities more dependent on reservoir hyperparameters. Across both sign-balance and shuffle controls, this tradeoff is closely tied to the raw spectral radius before normalization. Perturbations that lower the raw spectral radius are rescaled more strongly to the target radius and tend to increase both performance and CV, whereas perturbations that raise the raw spectral radius tend to decrease both. Thus, the biological sign structure is not interpreted as an optimum for reservoir performance; instead, it participates in a broader structural regime that favors lower hyperparameter sensitivity by shaping the spectrum of the recurrent matrix before normalization.

1. Introduction

Reservoir computing uses a fixed recurrent dynamical system to project input histories into a high-dimensional state space, after which only a readout is trained (Jaeger, 2001; Maass, Natschläger and Markram, 2002). This separation makes reservoir models useful for studying how recurrent architecture shapes computation. It also exposes a persistent design problem: reservoir behavior can depend strongly on hyperparameters such as spectral radius, input scaling, leak rate, and neuronal bias. A reservoir that performs well only under a narrow set of choices is less useful as a robust computational substrate.

Most reservoir-computing theory and practice treats random recurrent networks as a strong generic baseline. Random reservoirs can provide diverse, weakly correlated nonlinear features and can be tuned near dynamical regimes that support memory and separability (Lukoševičius, 2012). From that perspective, a small, sparse, structured, signed biological circuit should not be expected to maximize generic task-agnostic capacity. The *C. elegans* connectome

ORCID(s):

was shaped by developmental, metabolic, behavioral, and stability constraints, not by the objective of serving as an optimized random feature map. Its value here is therefore not as a presumed best reservoir, but as a structured biological prior for asking which constraints change the relationship between performance and sensitivity.

This distinction matters because biological neural systems often remain functional across variable internal and environmental conditions, whereas artificial systems can be brittle outside the regime in which they were tuned or trained (Recht, Roelofs, Schmidt and Shankar, 2019; Geirhos, Jacobsen, Michaelis, Zemel, Brendel, Bethge and Wichmann, 2020). In reservoir computing terms, this suggests a question that is different from peak-performance optimization: which structural properties make computational statistics stable across hyperparameter variation? The present study addresses that question by treating hyperparameter invariance as a metric of reservoir architectures.

We use the signed chemical-synapse *C. elegans* connectome as a biologically grounded reservoir and compare it with controlled perturbations that isolate topology, excitatory/inhibitory sign structure, weight-magnitude distribution, magnitude placement, and sign placement. Reservoir behavior is evaluated with four task-agnostic metrics: memory capacity (MC), information processing capacity (IPC), kernel rank (KR), and generalization rank (GR). Hyperparameter invariance is quantified as the coefficient of variation of these metrics across a full factorial sweep of spectral radius, input scaling, leak rate, and neuron-bias range.

The results identify a performance–invariance tradeoff rather than a biological optimum for reservoir performance. Perturbations that increase task-agnostic capacity often also increase sensitivity to hyperparameter choice. Global excitatory/inhibitory sign imbalance strongly influences the low-variance regime observed in the *C. elegans* model studied here, but it is not a universal determinant acting alone. Its effect depends on topology, local sign placement, weight-magnitude distribution, and magnitude placement because these structural features change the raw spectral radius that is subsequently normalized during reservoir evaluation.

2. Methods

2.1. Reservoir model

We use echo state networks, a discrete-time reservoir-computing model in which recurrent weights are fixed and only linear readouts are fit when a metric requires reconstruction (Jaeger, 2001; Lukoševičius, 2012). Let $u_t \in \mathbb{R}^d$ be the input, $x_t \in \mathbb{R}^N$ the reservoir state, and $y_t \in \mathbb{R}^p$ the readout. With leak rate $\alpha \in (0, 1]$, the state update is

$$\tilde{x}_{t+1} = \tanh(W^{\text{res}}x_t + s_{\text{in}}W^{\text{in}}u_{t+1} + b + \xi_t), \quad (1)$$

$$x_{t+1} = (1 - \alpha)x_t + \alpha \tilde{x}_{t+1}, \quad (2)$$

$$y_t = W^{\text{out}}x_t. \quad (3)$$

Here $W^{\text{res}} \in \mathbb{R}^{N \times N}$ is the sparse recurrent reservoir matrix, W^{in} is the input matrix, s_{in} scales the input drive, $b \in \mathbb{R}^N$ is a fixed per-neuron bias vector, and ξ_t denotes optional noise. Noise is not included in the experiments. The readout dimension is $p = 5$ for IPC so that multiple polynomial orders can be solved in a single batch; otherwise $p = 1$.

Only W^{out} is fit during metric calculations. The recurrent matrix is fixed after the architecture is instantiated, so the experiments isolate how reservoir structure and global hyperparameters shape the state matrix available to the readout. Architectures are rescaled to a target spectral radius before each trial is evaluated, as described below. This keeps the comparison focused on relative structure rather than raw matrix scale.

2.2. Reservoir hyperparameters

The experiments sweep four hyperparameters that control the reservoir's operating regime. The spectral radius $\rho(W^{\text{res}})$ controls the strength of recurrent feedback, where

$$\rho(W) = \max_{1 \leq i \leq N} |\lambda_i|.$$

This target is imposed after each architecture or perturbation has been instantiated. Let W' denote the signed recurrent matrix after all topology, sign, and magnitude manipulations have been applied. For a target spectral radius ρ_{target} , the matrix used during the trial is

$$W^{\text{res}} = \frac{\rho_{\text{target}}}{\rho(W')} W', \quad \rho(W') = \max_i |\lambda_i(W')|.$$

Thus, spectral-radius scaling is the final matrix operation before the reservoir is driven by the input sequence.

This normalization is important because otherwise architecture comparisons could be dominated by raw weight scale rather than by topology, sign balance, or magnitude placement. Matching every instantiated matrix to the same target spectral radius makes the hyperparameter sweep compare reservoirs at matched recurrent gain values. Normalization does not erase the effect of structure: topology and sign placement determine the raw spectrum of W' , and therefore determine the scale factor required to reach ρ_{target} . For example, sign-balanced matrices can cancel reinforcing loops and change the leading eigenvalue structure, while strongly sign-imbalanced matrices tend to preserve more coherent positive feedback. The resulting normalization factor changes the effective scale of recurrent weights used during trials.

The leak rate α controls the effective timescale of the reservoir by interpolating between the previous state and the newly computed activation. Input scaling s_{in} controls the strength of the external drive relative to recurrent feedback. The neuron-bias half-range β controls each neuron's baseline operating point: when $\beta = 0$, $b = 0$; when $\beta > 0$, each neuron is sampled independently as $b_i \sim \mathcal{U}(-\beta, \beta)$ and then held fixed during trials. Positive bias values raise a neuron's baseline activation tendency, while negative values lower it.

These parameters do not alter the trained readout directly; they change the dynamics of the reservoir whose stability is evaluated by the invariance analysis.

2.3. Connectome data and preprocessing

A connectome is a description of the neural wiring of an organism. It encodes which neurons are connected, the direction of each connection, and often additional biological metadata such as the number of synaptic terminals and the neurotransmitter associated with each connection.

In this work, we treat the connectome as the foundation for a graph: each neuron is represented as a node, and each synapse as a directed, weighted edge. Two *C. elegans* connectome sources are used. The main experiments use the signed chemical-synapse connectome derived from EleganSign (Fenyves, Szilágyi, Vassy, Söti and Csermely, 2020; EleganSign, 2026). We analyze two preprocessing versions of this dataset. In the signed-only version, unknown or complex-sign connections are removed, leaving 297 nodes and 1740 directed signed edges. In the augmented random-sign version, the 1864 unknown or complex-sign edges are retained and assigned signs randomly with a 4:1 excitatory-to-inhibitory ratio. The older c302-derived connectome (Gleeson, Lung, Grosu, Hasani and Larson, 2018) is retained only for the supplemental architecture visualizations and legacy schematic controls; it is not the source of the main results figures.

The signed-only graph contains 1740 directed chemical-synapse edges over 297 neurons after preprocessing. These signs are treated as the empirical excitatory/inhibitory partition for the invariance experiments. Throughout this paper, excitatory and inhibitory refer to signed connections, not to neuron classes: a 50% inhibitory condition means that 50% of modeled edges have negative sign, not that 50% of neurons are inhibitory. The augmented random-sign graph adds the 1864 unknown or complex-sign connections back into the topology and assigns each such edge a random sign under a 4:1 excitatory-to-inhibitory prior. This assignment is used as a sensitivity analysis rather than as a claim that those individual signs are known. Both versions remain *working abstractions*: they do not attempt to capture receptor-specific, context-dependent, developmental, or neuromodulatory effects. Because this study examines how coarse sign structure and weight statistics influence task-agnostic reservoir metrics across hyperparameter sweeps, incorporating detailed synapse-type dynamics is beyond the scope of the analysis and computationally expensive at the current scope of our experiments.

When edge magnitudes are required, the number of synaptic contacts from neuron i to neuron j is used as a weight proxy, $W_{ij} = c_{ij}$. Treating contact number as a proxy weight is standard in *C. elegans* modeling (Gleeson et al., 2018; Casal, Galella, Vilarroya and Garcia-Ojalvo, 2020; Churchland and Garcia-Ojalvo, 2026) and provides the biologically informed magnitude prior for the reservoir.

2.4. Task-agnostic reservoir metrics

Task-agnostic reservoir statistics measure computational properties of the reservoir before training it on a specific downstream task. This study uses four such metrics: kernel rank (KR), generalization rank (GR), memory capacity (MC), and information processing capacity (IPC), following prior work on reservoir/neural-microcircuit diagnostics and benchmarking (Maass, Legenstein and Bertschinger, 2004; Jaeger, 2002; Dambre, Verstraeten, Schrauwen and Massar, 2012; Pilati, Ceni, Michieletti, Gallicchio, Ricciardi and Milano, 2026). For each architecture and hyperparameter setting, the reservoir is driven by an i.i.d. input sequence and the resulting state matrix is used to compute these metrics. Higher MC, IPC, and KR indicate greater task-agnostic computational capability. GR measures sensitivity to small input perturbations; throughout the results it is analyzed alongside the other metrics as a reservoir response statistic.

2.4.1. Kernel Rank

Kernel rank measures how many effectively independent reservoir activity dimensions are produced by the input drive (Maass et al., 2004). We collect the post-washout state matrix $X \in \mathbb{R}^{T \times N}$, center it over time, compute singular values $S = \{\sigma_i\}$, normalize them as $p_i = \sigma_i / \sum_j \sigma_j$, and compute effective rank:

$$\text{KR} = \text{erank} = \exp \left(- \sum_i p_i \ln p_i \right).$$

This effective-rank formulation captures the spread of activity across orthogonal modes rather than exact linear independence alone (Roy and Vetterli, 2007).

2.4.2. Generalization Rank

Generalization rank measures how reservoir states change under small input perturbations (Maass et al., 2004). Let $u_t \sim \mathcal{U}(-1, 1)$ be the clean input and $u'_t = u_t + \kappa \varepsilon_t$, where $\varepsilon_t \sim \mathcal{N}(0, 1)$ and $\kappa = 0.01$. If X and X' are the corresponding post-washout state matrices, then

$$\text{GR} = \text{erank}(X' - X).$$

2.4.3. Memory Capacity

Memory capacity measures how well a linear readout can reconstruct past inputs from the current reservoir state (Jaeger, 2002). For each delay d , a ridge-regression readout is trained to predict $u(t - d)$ from $x(t)$, and the squared Pearson correlation between predicted and true delayed input is recorded. We approximate total memory capacity by

summing over delays:

$$\text{MC} = \sum_{d=1}^{D_{\max}} r_{\text{score}}(u(t-d), \hat{u}_d(t))^2.$$

We use $D_{\max} = 30$. Validation sweeps over $D_{\max} \in \{30, 50, 80\}$ showed that this cutoff preserved essentially the full measured MC in the models tested.

2.4.4. Information Processing Capacity

Information processing capacity measures how well the reservoir can reconstruct nonlinear functions of delayed inputs (Dambre et al., 2012). We use an incomplete, truncated single-delay IPC diagnostic rather than the full IPC decomposition. Specifically, we evaluate univariate Legendre polynomial targets of one delayed input at a time and omit pairwise and higher-order cross-delay product terms. This still measures the reservoir’s ability to generate nonlinear transformations of input history, but it does not estimate the full capacity over all nonlinear input combinations. For Legendre polynomial orders $K = \{1, 2, 3, 4, 5\}$, let $z_{k,d}(t) = P_k(u(t-d))$. We compute

$$\text{IPC} = \sum_{d=1}^{D_{\max}} \sum_{k \in K} \max\left(0, r_{\text{score}}(z_{k,d}(t), \hat{z}_{k,d}(t))^2\right).$$

We use $D_{\max} = 30$. Validation against longer delay windows and univariate polynomial orders 1 through 10 showed that this incomplete single-delay IPC closely matched the larger single-delay computation while reducing runtime. Because cross-delay products are not included, the reported IPC values should be interpreted as an incomplete IPC-style statistic for nonlinear input transformations rather than as an exhaustive estimate of full IPC.

2.5. Hyperparameter-invariance metric

We quantify hyperparameter invariance by measuring how much each task-agnostic metric changes across the hyperparameter grid. Because MC, IPC, KR, and GR have different numerical scales, we use the coefficient of variation (CV) rather than raw variance:

$$CV = \frac{\text{std}(a)}{\bar{a}}$$

where a is the set of values for a metric across the evaluated hyperparameter settings within a trial. Low CV indicates that a metric is stable under hyperparameter variation; it does not by itself imply high absolute performance. The results therefore analyze CV together with mean metric value.

Table 1

Architecture families used in the reservoir ablations.

Architecture family	Manipulation	Main comparison
Empirical baseline	Original <i>C. elegans</i> topology, signs, and contact-count magnitudes	Defines the low-variance reference regime
Sign-preserving simplifications	Replace magnitudes with signed unit, half-Gaussian, or uniform magnitudes while preserving the empirical sign ratio	Tests how much invariance is recovered by preserving global E/I sign imbalance
Binary weights	Collapse all nonzero weights to positive binary values	Tests the effect of removing inhibitory signs and weight magnitudes
Magnitude-placement controls	Shuffle empirical magnitudes across fixed edges or interpolate toward shuffled magnitudes while preserving signs	Tests whether empirical magnitudes matter through their placement
Degree-preserving topology controls	Rewire edges using directed degree-preserving connection shuffles, with or without magnitude shuffling	Tests topology and its interaction with heterogeneous weights
Random-topology baselines	Use Gaussian-weight <i>C. elegans</i> -topology, ER, and WS controls	Tests whether topology alone explains the low-variance regime

2.6. Structural perturbations and architecture controls

To determine which topological and weight-related features influence robustness, we construct a series of controlled network variants derived from the *C. elegans* connectome. Each architecture is designed to isolate a specific property of the biological network—such as topology, weight distribution, sign structure, or magnitude placement—while holding other properties fixed. Table 1 summarizes the architecture families used in the ablations; full architecture definitions and visualizations are provided in Supplemental Section A.3. The procedural operators used to instantiate these controls are defined below.

The small-world model belongs to the random-topology baseline family rather than to the perturbation operators below. Small-world networks combine relatively high clustering with short path lengths (Barabási, 2016). They are included because prior reservoir-computing work found that small-world reservoirs can maintain memory performance across a broader range of spectral radii than fully random reservoirs when input and output nodes are spatially segregated (Kawai, Park and Asada, 2019; Watts and Strogatz, 1998). In this study, the small-world network is used only as a topology control: it tests whether a generic clustered-and-short-path architecture reproduces the invariance profile of the *C. elegans* reservoir once empirical signs and magnitudes are removed.

2.6.1. Connection shuffle

We define a connection shuffle as a directed degree-preserving edge swap. Let W be the weighted adjacency matrix. We randomly select two existing directed edges (a, b) and (c, d) , where a, b, c, d are distinct and the proposed edges

(a, d) and (c, b) do not already exist. We then form a new matrix W' by setting

$$W'_{ad} = W_{ab}, \quad W'_{cb} = W_{cd}, \quad W'_{ab} = 0, \quad W'_{cd} = 0,$$

with all other entries unchanged. In this operation, the weights are carried with the original edges as they are reassigned to new targets. This preserves each node's in-degree and out-degree while shuffling the neighborhood structure.

2.6.2. Weight shuffle

Fix the synaptic connectivity pattern and randomly permute the weights assigned to existing connections. Let

$$NZ = \{(i, j) : W_{ij} \neq 0\}$$

denote the set of nonzero entries in the weight matrix. We extract the corresponding weights into a vector,

$$w_{NZ} = \{W_{ij} : (i, j) \in NZ\},$$

randomly shuffle this vector using `numpy.random.shuffle`, and then reassign the shuffled weights back to the same nonzero indices. Thus, the set of existing connections is unchanged, but the weights assigned to those connections are redistributed across the fixed network topology.

For this shuffle, the probability that a given connection changes sign depends on the empirical sign ratio of the graph being shuffled. This operation is therefore not sign-preserving unless explicitly stated otherwise.

2.6.3. Sign-preserving magnitude interpolation

To separate the effect of weight magnitudes from the effect of excitatory/inhibitory sign balance, we also use sign-preserving magnitude interpolation. These experiments keep the biological topological and synapse signs fixed. Only the nonzero magnitudes are changed.

Let

$$s_{ij} = \text{sign}(W_{ij}^{\text{cel}})$$

for each nonzero *C. elegans* edge. Given two nonnegative magnitude assignments m_{ij}^0 and m_{ij}^1 defined on the same edge set, the interpolated weight matrix at fraction $f \in [0, 1]$ is

$$W_{ij}^f = s_{ij} \left((1 - f)m_{ij}^0 + fm_{ij}^1 \right).$$

Thus, the sign of every edge is identical for all interpolation fractions. This differs from the weight-shuffle procedure in Section 2.6.2: in the interpolation experiments, shuffled *C. elegans* magnitudes are permuted as magnitudes only and are then assigned to edges whose original signs are preserved.

We evaluate three interpolation families. First, we interpolate from signed unit magnitudes to empirical *C. elegans* magnitudes:

$$m_{ij}^0 = 1, \quad m_{ij}^1 = |W_{ij}^{\text{cel}}|.$$

Second, we interpolate from signed unit magnitudes to shuffled empirical magnitudes:

$$m_{ij}^0 = 1, \quad m_{ij}^{f1} = m_{ij}^{\pi},$$

where m^{π} is a random permutation of the empirical magnitudes $|W^{\text{cel}}|$ across the nonzero edge set. Third, we interpolate directly from empirical magnitudes to shuffled empirical magnitudes:

$$m_{ij}^0 = |W_{ij}^{\text{cel}}|, \quad m_{ij}^1 = m_{ij}^{\pi}.$$

In all three cases, topology and signs are fixed, so changes in CV or mean metric value reflect magnitude perturbation rather than sign perturbation.

2.7. Trial definition

A single trial of a model consists of:

1. A reservoir instantiation for each trial, reset between hyperparameter combinations
2. A performance evaluation for each hyperparameter combination, using the following parameters:
 - **Sequence length** = 2500
 - **Washout time steps** = 500
 - **Perturbation standard deviation** = 0.01 (for calculating GR)
 - **Train time steps** = 1500
 - **Test time steps** = 500
 - **Memory capacity max delay** = 30
 - **Information processing capacity max delay** = 30

- **Information processing capacity polynomial targets** = univariate Legendre orders $\{1, 2, 3, 4, 5\}$; pairwise and cross-delay product terms are not included
- **Ridge alpha** = 0.0001 (controls the strength of L2 regularization for ridge regression)

2.8. Hyperparameter sweeps

We evaluated network performance across a full factorial sweep of spectral radius, leak rate, input scaling, and neuron-bias range. To compute our coefficient of variation (CV) metric, each trial evaluates a given network's performance across all combinations of the following hyperparameters:

- Spectral radius: $sr \in \{0.6, 0.8, 0.95, 1.05\}$
- Leak rate: $leak \in \{0.6, 0.8, 1.0\}$
- Input scaling: $input\ scaling \in \{0.1, 0.5, 1.0, 1.5\}$
- Neuron bias half-range: $\beta \in \{0.0, 0.1\}$

The coefficient of variation (CV) is computed across hyperparameter combinations within a trial. Each trial uses a single fixed input sequence shared across all hyperparameter combinations to ensure comparability. Reservoir instantiations are independent across trials: for each trial, the reservoir is reconstructed with a unique seed. Each reservoir instantiation in a trial uses the same input sequence; different trials use different input sequences.

This factorial design produces 96 hyperparameter settings per trial. Number of trials for architecture comparison 1: 200 per model. Total metric evaluations per model: Trials $\times$ hyperparameter grid = 19,200. Number of trials for Figures 2, 6, and 7: 50 per point. Total metric evaluations per point: 4,800. The half-normal magnitude-perturbation control also uses 50 trials per point, for 4,800 total metric evaluations per point. The interpolation experiments in Figures 13, 14, and 15 use the same hyperparameter grid and compute CV independently at each interpolation fraction $f \in \{0, 0.25, 0.5, 0.75, 1\}$.

2.9. Experimental pipeline

For each trial:

1. Instantiate reservoir computer (nodes and edges)
2. Assign weights/signs
3. Scale to target spectral radius
4. Create an i.i.d. input sequence
5. Run the network on the input sequence and split states into washout/train/test sets

6. Compute task-agnostic metrics (KR, GR, IPC, MC)
7. Repeat for each combination of hyperparameters in our hyperparameter array

Analysis of Trials:

1. Compute CV across hyperparameter grid
2. Average over trials
3. Analyze results

3. Results

C. elegans reservoirs are relatively insensitive to reservoir hyperparameter choices, but the experiments below do not support a flat list of equally important or independent causes. In the present model, global excitatory/inhibitory (E/I) sign imbalance is the strongest single control we identify: preserving the empirical sign ratio recovers much of the connectome’s invariance for IPC, KR, and GR, even when exact sign placement or empirical magnitudes are altered or simplified. This should not be read as a universal claim about E/I balance alone. The relative effect of sign imbalance depends on the surrounding architecture, including topology, sign placement, weight-magnitude distribution, and magnitude placement.

3.1. The empirical connectome occupies a low-variance reservoir regime

Figure 1 establishes the basic phenomenon. Across sweeps of spectral radius, leak rate, input scaling, and neuron bias, the empirical *C. elegans* connectome lies in a low-CV regime for the four task-agnostic metrics considered here. Low CV means that the metric changes relatively little as hyperparameters vary, so this regime corresponds to hyperparameter invariance rather than high absolute performance.

The architecture comparison gives an initial ordering of possible causes within this experimental family. Models that do not preserve the empirical sign structure, shown in yellow, occupy the highest-CV region and generally move toward higher mean performance. The maximally imbalanced binary-weight control occupies the lowest-CV, lowest-performance island. By contrast, sign-preserving models remain closer to the empirical connectome, even when their nonzero weight magnitudes are simplified or altered. This suggests that weight magnitudes and their placement contribute to invariance in the *C. elegans* reservoir model, but their effects are smaller than the effect of preserving the empirical E/I sign structure. The performance–CV island plot makes this ordering visible in the joint space of mean metric value and hyperparameter sensitivity: the empirical connectome occupies a compact low-CV island, while balanced random-sign controls move toward higher performance and higher CV. These comparisons motivate the central claim tested in the following sections: global E/I sign imbalance is the strongest contributor to the connectome’s

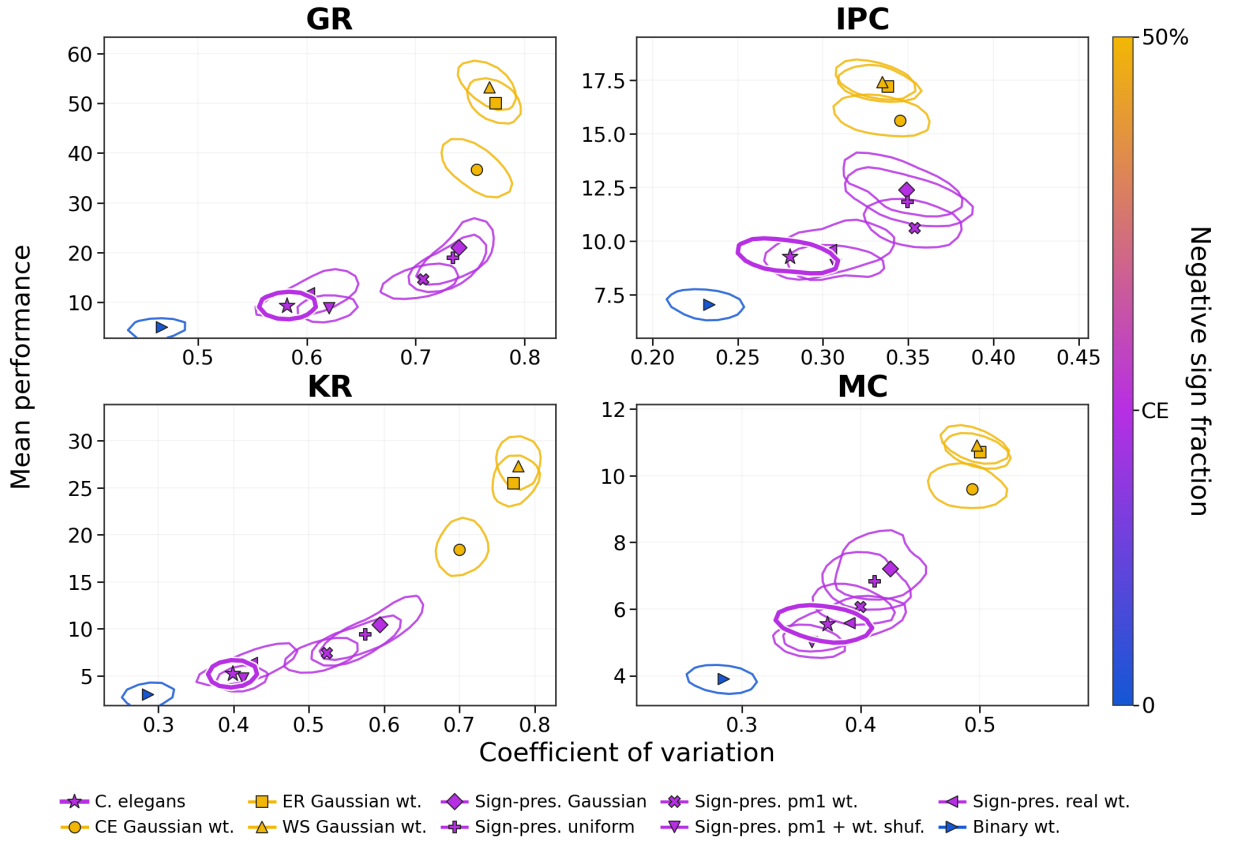


Figure 1: Performance–CV island plots for the architecture comparison using 50% density contours. Each panel shows mean performance against hyperparameter CV for one task-agnostic metric. Points mark condition centers; the black starred island marks the empirical *C. elegans* connectome.

low-variance behavior under the present topology and magnitude regimes, while sign placement and weight magnitudes have smaller effects.

3.2. Global sign balance reveals a performance–invariance tradeoff

To isolate global sign balance, we flipped the signs of randomly selected synapses while preserving the remaining structure of each network. This manipulation changes the inhibitory edge fraction while holding connectivity and magnitudes fixed. It does not assign neurons to excitatory or inhibitory classes. The result is a clear performance–invariance tradeoff. In the all-positive or strongly sign-imbalanced regime, reservoirs have low CV but reduced mean computational performance. As the fraction of negative edges increases toward a more balanced E/I edge-sign regime, mean metric values generally rise, but CV rises as well (Figure 2). The same qualitative trend appears in the *C. elegans*-matched ER topology and in the unknown/complex-removed *C. elegans* magnitude pattern (Supplemental

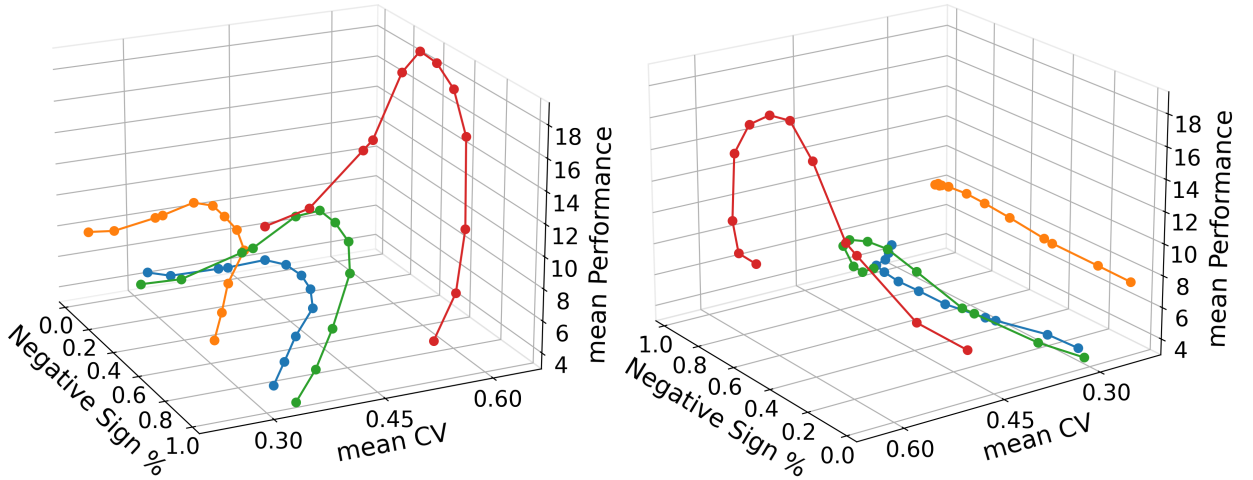


Figure 2: Mean performance and mean hyperparameter CV as a function of negative sign percentage for the *C. elegans* topology with real weights, using the augmented graph in which complex or unknown neurotransmitter signs are assigned under a 4:1 excitatory-to-inhibitory prior. IPC is denoted in orange, GR in red, MC in blue, and KR in green. Increasing the negative sign percentage moves the reservoir from a low-variance, low-performance regime toward a higher-performance but more hyperparameter-sensitive regime. A total of 50 trials were performed for each fraction.

Figures 6 and 7). Thus, the same manipulation that increases computational capability also makes the reservoir more sensitive to spectral radius, leak rate, input scaling, and neuron bias.

This performance trend is consistent with recent reservoir studies in which balanced or slightly inhibition-dominated regimes outperform excitation-dominated regimes on memory, prediction, and locomotor-pattern tasks (Srinivasan, Plenz and Girvan, 2025; de Graaf, Wagner, Mochizuki and Le Mouel, 2025). The present result extends that framing by emphasizing not only mean performance, but also the accompanying increase in hyperparameter sensitivity.

This pattern explains why preserving the empirical sign ratio recovers much of the connectome’s invariance in the architecture comparison. The empirical connectome, under the coarse neurotransmitter-to-sign mapping used here, sits on the low-variance side of the tradeoff rather than near the peak-performance region. In the reservoir-computing abstraction studied here, the approximately 4:1 excitatory-to-inhibitory balance associated with many biological networks does not prioritize peak task-agnostic performance. Instead, it places the reservoir in a lower-performance but more hyperparameter-invariant regime. Thus global E/I sign imbalance is a strong control parameter in this model, but its effect is expressed through the tested topology and magnitude regimes.

3.3. Sign-normalization links sign balance to spectral-radius scaling

The sign-fraction curves can be interpreted through the way edge-sign balance changes the raw spectral radius before normalization. Changing the fraction of negative edges changes whether recurrent loops reinforce each other or cancel. More negative edges can reduce the raw spectral radius by disrupting coherent positive feedback. When each perturbed matrix is normalized by its own spectral radius, $W/\rho(W)$, a lower raw $\rho(W)$ produces a larger scale factor

for the same target spectral radius. In other words, sign balance can indirectly allow larger effective recurrent weights after normalization.

The sign-normalization ablations test this mechanism by comparing the usual own-radius normalization, $W/\rho(W)$, with normalization by the original matrix's spectral radius, $W/\rho(W_{\text{orig}})$. Under own-radius normalization, performance increases strongly as the negative edge fraction moves toward the balanced regime, and CV increases at the same time (Figure 3). When the same perturbed matrices are instead scaled by the original spectral radius, the performance curves are much flatter through the middle of the sign-fraction sweep. The ER-matched and unknown/complex-removed controls show the same qualitative separation between normalization conventions (Supplemental Figures 9 and 10).

This flattening is expected only over the range of sign fractions where the perturbed matrix has a lower raw spectral radius than the original matrix. Let q denote the negative edge fraction and q_{orig} the original fraction. For the sign-sweep experiments, this middle region is approximately the interval between q_{orig} and its sign-complement $1 - q_{\text{orig}}$. In that range, $\rho(W) < \rho(W_{\text{orig}})$, so original-radius scaling gives a post-scaling radius

$$\rho_{\text{post}} = \rho_{\text{target}} \frac{\rho(W)}{\rho(W_{\text{orig}})} < \rho_{\text{target}}.$$

Though the weight magnitudes are not changing, the post-normalization spectral radius is. In all three experiments, performance stays flat in these inner fractions. Outside this middle range, the perturbed matrix can have $\rho(W) > \rho(W_{\text{orig}})$. Scaling by $\rho(W_{\text{orig}})$ then moves the post-scaling spectral radius back toward, or above, the target value rather than holding it low. The changeover occurs where $\rho(W) = \rho(W_{\text{orig}})$.

In the *C. elegans* sign sweeps, this changeover can occur slightly before the empirical negative-edge fraction rather than exactly at it. The reason is that the sweep does not only change the global fraction of negative edges; it also randomly reassigns which edges carry those signs. The shuffle controls show that randomizing sign placement can itself increase performance and CV, consistent with a reduction in ρ_{raw} (Figure 4). A quick comparison of the sign sweeps in the *C. elegans* and ER-matched networks is given in Supplemental Figure 8. Therefore, the perturbed *C. elegans* matrix can reach $\rho(W) < \rho(W_{\text{orig}})$ before the sweep reaches the empirical sign fraction. This offset is not expected to appear as strongly in the ER-matched network, where signs are already effectively random with respect to the topology and reshuffling sign placement has little additional effect on the raw spectral radius.

This pattern suggests that the performance gain from sign balance is mostly mediated by spectral-radius scaling. The benefit is not only that positive and negative edges create a different signed topology; it is also that this signed topology lowers the raw spectral radius, so own-radius normalization applies a larger scale factor. That higher effective recurrent scale improves task-agnostic performance but also makes the reservoir more hyperparameter-sensitive. This interpretation is consistent with recent reservoir-computing work on the no-strong-loops principle: Hadaeghi

and colleagues show that stronger reciprocal loops inflate $\rho(W)$, forcing spectral-radius normalization to downscale recurrent weights more strongly and reducing memory capacity and kernel rank (Hadaeghi, Fakhar, Khajehnejad and Hilgetag, 2025). Their manipulation targets reciprocity rather than E/I edge signs, but the mechanism is complementary. Strong loops raise the raw spectral radius and suppress effective recurrent scale after normalization; balanced edge signs can lower the raw spectral radius and permit larger effective recurrent scale.

3.4. Shuffle controls implicate raw spectral radius as a common mediator

The sign-balance result explains the main architecture-level separation, but it does not explain differences among controls that share the same global sign ratio. The degree-preserving shuffle experiments show that these residual differences follow the same spectral-radius mechanism observed in the sign-fraction experiments. Let ρ_{raw} denote the spectral radius of the perturbed matrix before it is rescaled to the target value. Across the shuffle controls, decreases in ρ_{raw} are associated with movement toward higher mean performance and higher CV, whereas increases in ρ_{raw} are associated with movement toward lower mean performance and lower CV.

With the original real-valued *C. elegans* weights, weight shuffling, connection shuffling, and connection-plus-weight shuffling all move the reservoir away from the unshuffled empirical connectome toward higher mean performance and higher CV (Figure 4, bottom row). The color-coded contours show that these controls also have lower raw spectral radius than the empirical baseline. This is the same direction seen in the sign-balance ablation: lowering ρ_{raw} makes the own-radius normalization apply a larger scale factor, increasing the effective recurrent weights after normalization and producing the performance–invariance tradeoff described above.

The ordering of the real-weight shuffle controls is still informative. In a connection shuffle, weights move with their original outgoing edge source as edges are reassigned to new targets. In a weight shuffle, magnitudes are instead redistributed across the fixed empirical edge set, so source-specific patterns of outgoing synaptic strength are broken. The connection-plus-weight shuffle first disrupts the neighborhood structure and then breaks this source-associated magnitude pattern. Its lower performance and CV relative to the pure connection shuffle indicate that the implicit weight movement caused by connection shuffling is not equivalent to an unconstrained weight shuffle. Keeping heterogeneous weights tied to their original source neuron produces a reservoir with lower ρ_{raw} than randomizing those magnitudes independently. The smaller differences in CV, performance, and spectral radius between the weight-shuffle and connection-plus-weight-shuffle controls suggest that, within this real-weight setting, the effect of topology alone is limited.

When all nonzero weights are collapsed to ± 1 while preserving the empirical sign ratio, the direction reverses (Figure 4, middle row). Using the original local sign placement as the baseline, the sign- and topology-shuffled signed-unit controls have lower CV across all four metrics (Supplemental Table 3). These controls also have higher ρ_{raw} ,

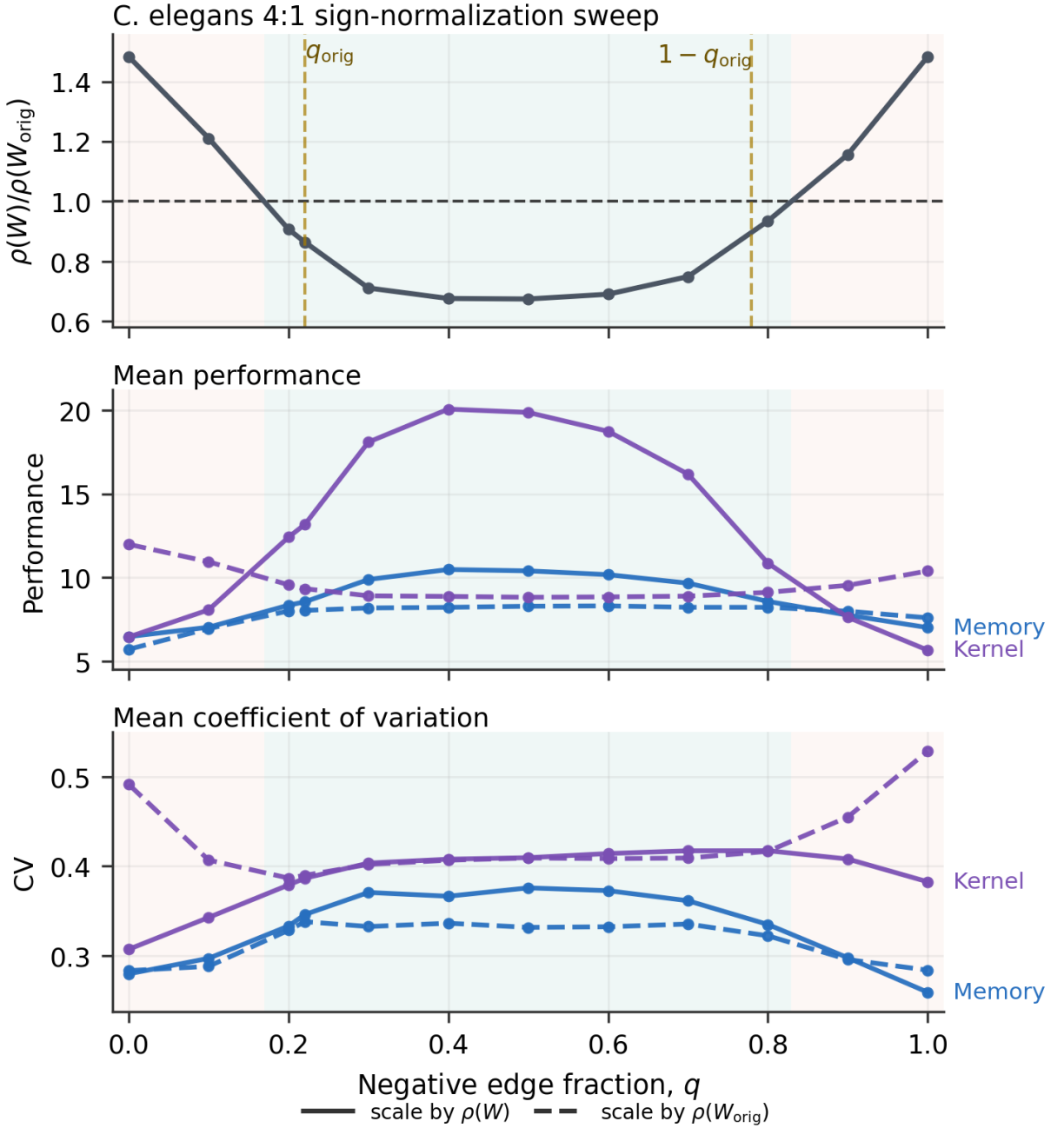


Figure 3: C. elegans 4:1 sign-normalization sweep. Solid curves scale by $\rho(W)$, and dashed curves scale by $\rho(W_{\text{orig}})$. Own-radius normalization produces a stronger performance increase and a larger CV increase as the negative edge fraction approaches the balanced regime.

and hence lower performance and CV than the signed-unit baseline. Thus, preserving only the global sign ratio recovers a lower-variance regime, but it does not make local sign placement irrelevant: the original local sign arrangement lowers

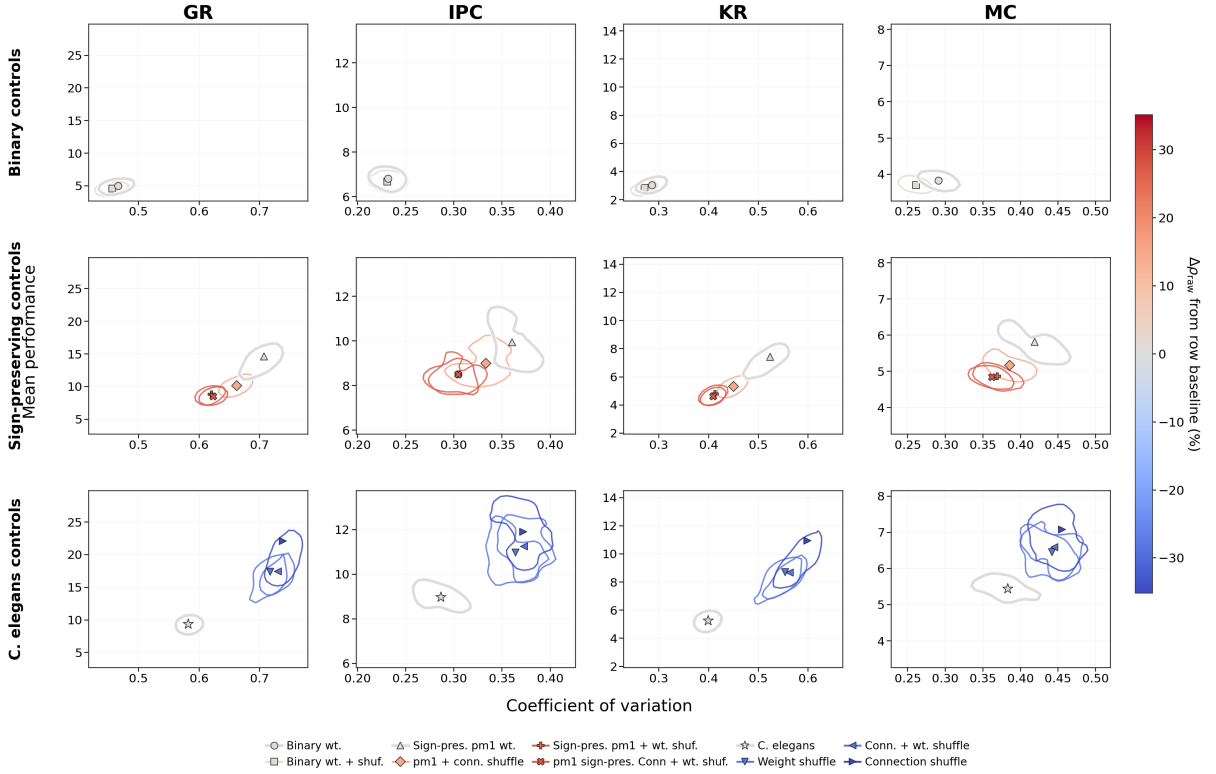


Figure 4: Performance–CV islands for the shuffle controls, shown with 50% density contours. Columns correspond to GR, IPC, KR, and MC. Rows show all-positive binary controls, sign-preserving signed-unit controls, and real-valued *C. elegans* controls. Contour color shows the percent change in raw spectral radius relative to the row baseline, $\Delta\rho_{\text{raw}}$; blue indicates lower raw spectral radius and red indicates higher raw spectral radius.

the raw spectral radius, placing the signed-unit reservoir in a higher-performance, higher-CV regime than the shuffled signed-unit controls.

The binary-weight experiment isolates topology further. Once every nonzero edge has the same positive sign and magnitude, degree-preserving rewiring only weakly changes ρ_{raw} and correspondingly produces smaller shifts in the performance–CV plane (Figure 4, top row; Supplemental Table 4). This supports the interpretation that topology matters, but that its effect on the tradeoff is strongest when it changes the raw spectrum of the recurrent matrix.

Together, the shuffle experiments make the hierarchy more mechanistic. Global E/I sign imbalance, local sign placement, topology, and heterogeneous magnitude placement are not independent explanations for the performance–invariance tradeoff. They are interdependent structural manipulations that shift ρ_{raw} before normalization. The direction of movement in the performance–CV plane is then largely predicted by whether that raw spectral radius decreases or increases relative to the relevant baseline.

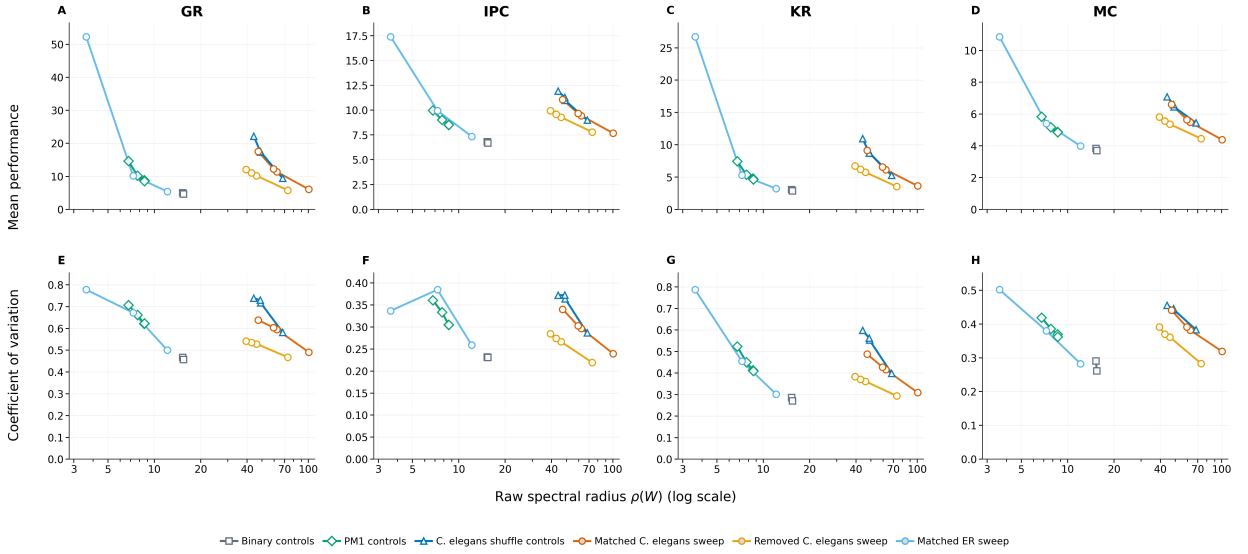


Figure 5: Relationship between raw spectral radius and reservoir behavior across sign-sweep and shuffle-control families. The top row shows mean task-agnostic performance as a function of raw spectral radius before normalization, ρ_{raw} , on a log scale. The bottom row shows the corresponding hyperparameter CV. Columns correspond to GR, IPC, KR, and MC. Lines connect points within experimental families and are not global fitted curves. Across families, lower ρ_{raw} generally coincides with higher performance and higher CV, while higher ρ_{raw} coincides with lower performance and lower CV.

3.5. Raw spectral radius summarizes the performance–invariance tradeoff

The sign-normalization and shuffle results point to the same mechanism: structural perturbations change the raw recurrent spectrum before spectral-radius normalization, and this changes the effective recurrent scale after normalization. Figure 5 summarizes this relationship across the sign-fraction sweeps and the shuffle controls. Across metrics, lower ρ_{raw} is generally associated with higher mean performance and higher CV, while higher ρ_{raw} is associated with lower performance and lower CV.

This summary should not be read as saying that raw spectral radius is an independent architectural feature separate from topology, sign placement, or weight magnitudes. Rather, those structural features appear to alter performance by changing ρ_{raw} . The plotted families show that the same direction appears in multiple experimental settings: balanced sign sweeps, signed-unit controls, binary topology controls, and real-weight shuffles. Thus, the final synthesis of the results is not simply that E/I balance matters, but that E/I balance, topology, and magnitude placement modulate a shared spectral-radius pathway that links reservoir architecture to the performance–invariance tradeoff.

4. Discussion

We find a hyperparameter invariance–performance tradeoff dictated by how network structure changes the raw recurrent spectrum before normalization. The *C. elegans* connectome occupies a relatively low-variance reservoir

regime, and the ablations point to an interwoven hierarchy of causes. Global E/I sign imbalance is the strongest structural perturbation in the model studied here: preserving the empirical sign ratio recovers much of the connectome’s invariance even when exact sign placement and empirical weight magnitudes are altered. However, the sign-normalization and shuffle results suggest that the more general mechanism is spectral. Sign balance, local sign placement, topology, and magnitude placement change ρ_{raw} ; spectral-radius normalization then converts those raw spectral differences into different effective recurrent weight scales.

Moving toward a more balanced E/I regime generally increases mean task-agnostic performance, but it also increases CV across spectral radius, leak rate, input scaling, and neuron bias. The same pattern appears in the shuffle controls: manipulations that lower ρ_{raw} tend to move reservoirs toward higher performance and higher CV, while manipulations that raise ρ_{raw} tend to move them toward lower performance and lower CV. Under the present modeling choices, the empirical *C. elegans* instantiation lies on the low-variance side of this curve rather than at the peak-performance region. Put more directly, the approximately 4:1 excitatory-to-inhibitory balance associated with many biological networks does not prioritize peak reservoir performance in this abstraction; it helps produce a more hyperparameter-invariant, lower-performance operating point by maintaining a raw spectrum that does not require as much post-normalization amplification.

4.1. Shape of the sign-balance curves

The approximate symmetry of the sign-fraction curves around a balanced edge-sign ratio is also consistent with the mathematical form of the reservoir used here. The state nonlinearity is $\tanh$, which is odd-symmetric, and the readout is linear. Under these choices, exchanging positive recurrent weights for negative recurrent weights changes the sign structure of the recurrent drive, but the linear readout can absorb corresponding sign changes in the state features. This makes the sign-flip experiment approximately symmetric. The centered peak near 50:50 edge signs is therefore consistent with balanced signs suppressing coherent recurrent feedback loops, lowering ρ_{raw} , and increasing the post-normalization effective recurrent scale. A centered peak in performance and a corresponding increase in CV near balanced edge signs are natural for this abstraction.

This helps explain why the present curves differ from studies in which the best operating point is slightly inhibition-dominated. Srinivasan, Plenz, and Girvan report performance benefits from balanced or slightly inhibition-dominated adaptive E/I regimes, while de Graaf and colleagues use a rectified $\tanh$ -like activation in a locomotor circuit model (Srinivasan et al., 2025; de Graaf et al., 2025). In such models, the activation function is not fully sign-symmetric: positive and negative inputs are transformed differently, so the optimal E/I ratio can shift away from exactly 50:50. By contrast, the odd activation and linear readout used here make the sign-flipping experiment more symmetric by

construction. The centered shape of the curves should therefore be interpreted as a property of this reservoir abstraction, not as a claim that biological circuits should generally optimize at equal excitation and inhibition.

Direct biological interpretation should be cautious because the connectome model compresses substantial uncertainty into a static signed network. The main analyses use EleganSign-derived signed connectomes, but many biological connections have unknown or complex sign annotations, and the inferred inhibitory edge fraction depends on how those ambiguous cases are handled (Fenyves et al., 2020; EleganSign, 2026). We therefore analyze both a signed-only version that removes those ambiguous edges and an augmented version that assigns them random signs under a 4:1 excitatory-to-inhibitory edge prior. If the true inhibitory edge fraction differs from either approximation, the empirical connectome may lie elsewhere along the observed performance–CV curve. In addition, E/I balance varies across development and aging (Wirak, Florman, Alkema, Connor and Gabel, 2022; Tanis, Bellemer, Moresco, Forbush and Koelle, 2009), and our model delivers inputs to all neurons rather than to biologically separated populations—such as sending inputs to input neurons and taking outputs from motor neurons. The present work should therefore be read as a structural reservoir analysis, not a faithful dynamical simulation of the animal.

For reservoir computing, the practical implication is that raw spectral radius before normalization should be treated as a key mediator between architecture and performance–invariance behavior. E/I sign ratio remains an important design parameter, complementing recent performance-focused work on adaptive E/I control in reservoirs (Srinivasan et al., 2025), but its effect should be evaluated through the spectrum it induces together with topology, sign placement, magnitude distribution, and magnitude placement.

4.2. Future directions

Several extensions would help clarify these interpretations. First, the model comparisons could be strengthened by adding additional topology controls that match degree and sign statistics more tightly, including degree-matched random and scale-free networks. Such controls would help determine which residual effects are genuinely topological and whether the distinctive sensitivity of MC persists outside the *C. elegans* degree distribution.

Second, the present conclusions should be tested on downstream tasks such as NARMA to determine whether the same structural relationships hold when reservoirs are evaluated by task performance rather than task-agnostic statistics alone. The same framework could also be used to investigate how connectome optimization algorithms (Churchland and Garcia-Ojalvo, 2026) alter task-agnostic statistics, and whether task-driven optimization moves biological reservoirs along the same performance–invariance axes identified here. Extending the ablation grid across additional topologies, magnitude distributions, and sign-placement rules would also test how far the relative importance of E/I imbalance generalizes beyond the present connectome-centered setting.

Third, it would be valuable to repeat these experiments with biologically separated input and output populations. Using sensory neurons as inputs and muscles or motor neurons as outputs would bring the model closer to a functional circuit interpretation and would better connect this work to prior studies of connectome-based computation (Kawai et al., 2019; Yu, Qin, Liu, Xu, Vogelstein, Brown and Vogelstein, 2025). Finally, the role of sign inference deserves more direct study. Natural next steps would be to compare alternative neurotransmitter-to-sign mappings and to repeat the sign-balance analyses under inferred inhibitory fractions closer to those suggested by recent biological work.

AI Usage Disclosure

Artificial intelligence tools, specifically ChatGPT (OpenAI, GPT-5 series) and Codex (OpenAI), were used as assistive tools during the course of this work.

AI tools were used to review text for clarity, grammar, and structure, and to identify ambiguous or overstated claims. All text included in this manuscript was written, reviewed, and edited by the authors, and any AI-assisted edits were manually verified and revised as necessary to ensure accuracy.

AI tools were also used to assist in identifying relevant literature. These tools were used in a search-assistance capacity to locate papers related to reservoir computing and recurrent biological networks. All cited papers were independently located, read, and verified by the author prior to inclusion.

In addition, AI tools were used as a programming reference to identify relevant libraries, functions, and debugging strategies. They were also used to assist in generating example plotting code and suggesting implementation approaches. All code used in this project was written, reviewed, and validated by the author.

AI tools were used solely as assistive instruments. All scientific claims, analysis, interpretations, and conclusions presented in this work are the independent work of the author.

A. Supplemental Information

A.1. Additional sign-fraction controls

The main text shows the sign-fraction sweep for the augmented *C. elegans* graph. Two additional controls show that the same performance–invariance tradeoff is not specific to that preprocessing choice.

A.2. Additional sign-normalization controls

The main text shows the sign-normalization ablation for the augmented *C. elegans* graph. The same normalization comparison was repeated for the matched ER topology and the unknown/complex-removed *C. elegans* magnitude pattern.

A.3. Architecture details

This section gives the full architecture definitions summarized in Table 1. The quantitative analyses use the EleanSign-derived preprocessing versions described in Section 2.3: a 297-node, 1740-edge signed-only graph and an augmented random-sign graph that retains the 1864 unknown or complex-sign edges under a 4:1 excitatory-to-inhibitory prior. The visualization panels in this supplemental section were generated from the earlier c302-derived connectome and are retained only as schematic illustrations of the control families.

A.3.1. *C. elegans connectome*

This uses the biological network and contact-count weights:

$$W_{ij} = W_{ij}^{cel}.$$

A.3.2. *Original C. elegans connectome with weight shuffle*

Starting from the original *C. elegans* connectome, this architecture preserves biological topology but shuffles the weights across existing connections using the weight-shuffle procedure in Section 2.6.2.

A.3.3. *Original C. elegans connectome with connection shuffle*

Starting from the original *C. elegans* connectome, this architecture applies the degree-preserving connection-shuffle procedure in Section 2.6.1.

A.3.4. *Connection shuffle with weight shuffle*

The *C. elegans* connectome is first connection-shuffled as described in Section 2.6.1, and the resulting nonzero weights are then shuffled as described in Section 2.6.2.

A.3.5. *C. elegans architecture with ± 1 weights using original connection signs*

This architecture preserves the biological topology and edge signs while setting every nonzero magnitude to one:

$$W'_{ij} = \begin{cases} \text{sign}(W_{ij}) & \text{if } W_{ij} \neq 0, \\ 0 & \text{if } W_{ij} = 0. \end{cases}$$

A.3.6. *C. elegans architecture with ± 1 weights and shuffled signs preserving sign balance*

This control keeps the biological connectivity fixed but randomizes which edges carry inhibitory signs. If

$$N_- = \#\{(i, j) : W'_{ij} = -1\},$$

then a random subset of size N_- of nonzero entries is assigned sign -1 , and all remaining nonzero entries are assigned sign $+1$. This preserves the global E/I sign ratio while shuffling sign placement.

A.3.7. *C. elegans architecture with ± 1 weights under connection shuffling*

Starting from the signed unit-magnitude network in Section A.3.5, the edge set is connection-shuffled as described in Section 2.6.1.

A.3.8. *C. elegans architecture with ± 1 weights under connection shuffling and weight shuffling*

Starting from the signed unit-magnitude network in Section A.3.5, the edge set is connection-shuffled and then the signed unit weights are shuffled across the resulting nonzero entries.

A.3.9. *C. elegans architecture with binary weights*

This ablation preserves the biological connectivity pattern but removes both sign and magnitude heterogeneity:

$$W'_{ij} = \begin{cases} 1 & \text{if } W_{ij} \neq 0, \\ 0 & \text{if } W_{ij} = 0. \end{cases}$$

A.3.10. *C. elegans architecture with binary weights and connection shuffle*

This binary network is rewired with the procedure in Section 2.6.1.

A.3.11. *C. elegans topology with Gaussian weights*

This control keeps the biological connectivity pattern but replaces each nonzero weight with an independent Gaussian sample:

$$W'_{ij} = \begin{cases} X_{ij} & \text{if } W_{ij} \neq 0, \\ 0 & \text{if } W_{ij} = 0, \end{cases} \quad X_{ij} \sim \mathcal{N}(0, 1).$$

A.3.12. *C. elegans–matched Erdős–Rényi model with Gaussian weights*

This random-graph baseline uses $N = 297$ nodes and an adjusted ER edge set with the same number of directed edges as the analysis *C. elegans* network. After fixing the edge count, weights are independently sampled from $\mathcal{N}(0, 1)$.

A.3.13. *Watts–Strogatz Gaussian control*

This small-world control uses $N = 297$ nodes and 1740 directed edges. A directed ring lattice is rewired with probability $p = 0.1$, preserving source out-degree and edge count, and nonzero weights are independently sampled from $\mathcal{N}(0, 1)$.

Table 2

Mean CV differences for real-valued shuffle controls relative to the empirical *C. elegans* reservoir. Baseline CV and control CV are mean coefficients of variation across the 96-point hyperparameter sweep. ΔCV is control minus baseline.

Metric	Control	<i>C. elegans</i> mean CV	Control mean CV	ΔCV	ΔCV (%)
MC	Conn. + wt. shuffle	0.3829	0.4448	+0.0620	+16.18
MC	Connection shuffle	0.3829	0.4549	+0.0721	+18.82
MC	Weight shuffle	0.3829	0.4418	+0.0589	+15.39
IPC	Conn. + wt. shuffle	0.2863	0.3724	+0.0861	+30.07
IPC	Connection shuffle	0.2863	0.3717	+0.0853	+29.80
IPC	Weight shuffle	0.2863	0.3638	+0.0775	+27.06
KR	Conn. + wt. shuffle	0.3985	0.5626	+0.1641	+41.17
KR	Connection shuffle	0.3985	0.5982	+0.1997	+50.11
KR	Weight shuffle	0.3985	0.5532	+0.1547	+38.80
GR	Conn. + wt. shuffle	0.5817	0.7296	+0.1479	+25.42
GR	Connection shuffle	0.5817	0.7381	+0.1564	+26.89
GR	Weight shuffle	0.5817	0.7172	+0.1354	+23.28

A.3.14. *C. elegans* architecture with half Gaussian weights using original connection signs

Starting from the signed unit-magnitude network in Section A.3.5, magnitudes are sampled from a half-Gaussian distribution:

$$W'_{ij} = \text{sign}(W_{ij})|\mathcal{N}(0, 1)|.$$

A.3.15. *C. elegans* architecture with uniform weights using original connection signs

Starting from the signed unit-magnitude network in Section A.3.5, magnitudes are sampled independently from a uniform distribution:

$$W'_{ij} = \text{sign}(W_{ij})X_{ij}, \quad X_{ij} \sim \mathcal{U}(0, 1).$$

A.4. Shuffle experiment CV summaries

Tables 2–4 report the paired changes in mean hyperparameter CV for the shuffle experiments shown in Figure 4. Positive values indicate higher CV than the corresponding baseline, and negative values indicate lower CV. Each comparison uses $n = 200$ paired trial groups.

A.5. Weight and magnitude ablation controls

The main results show that sign structure must be fixed before magnitude effects can be interpreted cleanly. The sign-preserving interpolation experiments therefore ask a narrower question: once topology and edge signs are held constant, does the placement of empirical magnitudes across signed edges matter?

Table 3

Mean CV differences for signed-unit shuffle controls relative to the original local sign-preserving signed-unit reservoir. Baseline CV and control CV are mean coefficients of variation across the 96-point hyperparameter sweep. ΔCV is control minus baseline.

Metric	Control	Sign-pres. pm1 wt. mean CV	Control mean CV	ΔCV	ΔCV (%)
MC	Sign-pres. pm1 + wt. shuf.	0.4186	0.3690	-0.0496	-11.84
MC	pm1 + conn. shuffle	0.4186	0.3855	-0.0331	-7.90
MC	pm1 sign-pres. + shuf.	0.4186	0.3617	-0.0569	-13.59
IPC	Sign-pres. pm1 + wt. shuf.	0.3604	0.3047	-0.0557	-15.46
IPC	pm1 + conn. shuffle	0.3604	0.3332	-0.0272	-7.54
IPC	pm1 sign-pres. + shuf.	0.3604	0.3045	-0.0559	-15.51
KR	Sign-pres. pm1 + wt. shuf.	0.5236	0.4126	-0.1110	-21.20
KR	pm1 + conn. shuffle	0.5236	0.4501	-0.0735	-14.04
KR	pm1 sign-pres. + shuf.	0.5236	0.4088	-0.1148	-21.92
GR	Sign-pres. pm1 + wt. shuf.	0.7066	0.6205	-0.0860	-12.18
GR	pm1 + conn. shuffle	0.7066	0.6618	-0.0448	-6.34
GR	pm1 sign-pres. + shuf.	0.7066	0.6232	-0.0833	-11.79

Table 4

Mean CV differences for the all-positive binary topology shuffle relative to the all-positive binary *C. elegans* topology. Baseline CV and control CV are mean coefficients of variation across the 96-point hyperparameter sweep. ΔCV is control minus baseline.

Metric	Control	Binary wt. mean CV	Control mean CV	ΔCV	ΔCV (%)
MC	Binary wt. + shuf.	0.2905	0.2610	-0.0295	-10.16
IPC	Binary wt. + shuf.	0.2317	0.2310	-0.0007	-0.30
KR	Binary wt. + shuf.	0.2862	0.2709	-0.0153	-5.34
GR	Binary wt. + shuf.	0.4668	0.4572	-0.0095	-2.04

Interpolating from signed unit magnitudes to the empirical *C. elegans* magnitudes produces only modest changes (Figure 13). At $f = 1$, CV changes by +7.39% for MC, -0.635% for IPC, +0.929% for KR, and +0.775% for GR. Mean metric values also change modestly. Thus, adding empirical magnitudes in their original locations does not substantially disrupt the low-variance regime once the empirical sign structure is already fixed.

The shuffled-magnitude interpolations behave differently. Moving from signed unit magnitudes to shuffled empirical magnitudes increases CV by 17.5% for MC, 14.3% for IPC, 32.1% for KR, and 24.4% for GR at $f = 1$ (Figure 14). Directly interpolating from empirical magnitudes to shuffled empirical magnitudes produces a similar pattern, with especially large effects for KR and GR (Figure 15). Because signs are preserved throughout these sweeps, the increase in CV cannot be attributed to altered E/I balance. It reflects the placement of heterogeneous magnitudes on the fixed signed topology.

The half-normal perturbation is best interpreted as a control rather than a main discovery. When connectivity and signs are preserved, moving the magnitude distribution toward a fitted shifted half-normal target produces only modest changes in both mean metric value and CV. When sign is not preserved in the corresponding control, the changes are

much larger. This confirms the ordering of causes: large effects attributed to magnitude perturbation can be dominated by sign-balance changes unless sign structure is explicitly fixed.

Together, the magnitude experiments complete the conditional hierarchy. Magnitude placement can move the reservoir toward higher mean performance and higher CV, particularly for KR and GR, but this effect appears after the sign structure has already set the main performance–invariance regime in these experiments. Empirical *C. elegans* magnitudes are therefore not irrelevant; they tune metric-specific behavior within a regime strongly shaped by global E/I sign imbalance.

A.6. Neuroscience Terms Explained for Computer Scientists

- **neuron** – basic building block of neural circuits. In this work, neurons are represented as nodes on a graph, each with both an activation function and an internal state representing membrane potential.
- **synapse** – another basic building block of neural circuits. In this work, synapses are represented by directed edges with weights connecting two neurons.
- **excitatory connection** – a modeled synaptic edge with positive sign
- **inhibitory connection** – a modeled synaptic edge with negative sign
- **E/I balance** – the ratio or fraction of excitatory to inhibitory synaptic signs in the modeled reservoir; in this paper this is an edge-level quantity, not a count of excitatory or inhibitory neurons
- **neurotransmitter** – a chemical that allows neurons to communicate with each other by transmitting signals across synapses (Sheffler, Reddy and Pillarisetty, 2025).
- **gap junction** – a type of synapse that connects the cytoplasm of two adjacent cells; bidirectional with the same weight for both sides. In our modeling, these are either removed or treated as two distinct positively signed edges.
- **neuromuscular junction** – a specialized synapse where a motor neuron communicates with a muscle fiber. These directed edges point to non-neural cells and hence are not included in our graph.
- **connectome** – a wiring diagram of a biological neural network. This typically contains all neurons and synapses, along with edge weights.
- **sensory neuron** – a neuron that receives sensory input. During typical biological use, these can be thought of as “inputs” to the network.
- **motor neuron** – a neuron that passes signals to motor cells. During typical biological use, these can be thought of as the “outputs” of the network.

Table 5

Changes in mean CV across Gaussian noise magnitudes. Baseline column lists the reference value for each metric; other entries show absolute change followed by percent change. Relative to baseline mode weight_test0.0. Sign is not preserved in this version of the experiment.

Metric	Baseline	$\Lambda = 1$	$\Lambda = 5$	$\Lambda = 10$
MC	0.3548	+0.00808 (+2.28%)	+0.0851 (+24%)	+0.162 (+45.6%)
IPC	0.2718	+0.00868 (+3.19%)	+0.0891 (+32.8%)	+0.151 (+55.4%)
KR	0.2634	+0.00505 (+1.92%)	+0.125 (+47.6%)	+0.335 (+127%)
GR	0.4426	+0.00829 (+1.87%)	+0.162 (+36.5%)	+0.311 (+70.3%)

Table 6

Changes in mean metric value across Gaussian noise magnitudes. Baseline column lists the reference value for each metric; other entries show absolute change followed by percent change. Relative to baseline mode weight_test0.0. Sign is not preserved in this version of the experiment.

Metric	Baseline	$\Lambda = 1$	$\Lambda = 5$	$\Lambda = 10$
MC	4.161	+0.00525 (+0.126%)	+0.715 (+17.2%)	+3.13 (+75.3%)
IPC	5.863	+0.0241 (+0.411%)	+0.999 (+17%)	+4.31 (+73.4%)
KR	2.99	+0.0545 (+1.82%)	+1.6 (+53.4%)	+7.51 (+251%)
GR	4.712	+0.12 (+2.55%)	+3.54 (+75%)	+16.4 (+348%)

- **synaptic terminals** – bulbous endings of a neuron’s axon where neurotransmitters are transferred to other cells (neurons or muscles). We use the number of these terminals as a proxy for edge weight.
- **membrane potential** – the difference in electrical charge between the inside and outside of a neuron, forming the basis for neural activity (Henley, 2021). For our purposes, this is the internal state of the neuron representing how much our activation function will be saturated.
- **small-worldness** – a property commonly observed in biological networks. A graph that exhibits the small-world property has both a high clustering coefficient and a short average path length. This structure is claimed to make information spread more efficient and to make the network more robust (Bassett and Bullmore, 2007; Latora and Marchiori, 2001).
- **scale-free network** – a network whose degree distribution follows a power law, meaning the fraction of nodes with degree k follows a power law $k^{-\alpha}$ (Broido and Clauset, 2019).

A.7. Tables

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Table 7

Changes in mean CV across Gaussian noise magnitudes. Baseline column lists the reference value for each metric; other entries show absolute change followed by percent change. Relative to baseline mode weight_test0.0. Sign is preserved in this version of the experiment.

Metric	Baseline	$\Lambda = 1$	$\Lambda = 5$	$\Lambda = 10$
MC	0.3623	+0.00232 (+0.641%)	+0.0133 (+3.67%)	+0.0156 (+4.32%)
IPC	0.2781	+0.00298 (+1.07%)	+0.0178 (+6.38%)	+0.0244 (+8.76%)
KR	0.2634	+0.00294 (+1.12%)	+0.0211 (+8.01%)	+0.0306 (+11.6%)
GR	0.4426	+0.00508 (+1.15%)	+0.0319 (+7.2%)	+0.0444 (+10%)

Table 8

Changes in mean metric value across Gaussian noise magnitudes. Baseline column lists the reference value for each metric; other entries show absolute change followed by percent change. Relative to baseline mode weight_test0.0. Sign is preserved in this version of the experiment.

Metric	Baseline	$\Lambda = 1$	$\Lambda = 5$	$\Lambda = 10$
MC	4.096	-0.00846 (-0.206%)	-0.094 (-2.3%)	-0.102 (-2.48%)
IPC	5.733	+0.00524 (+0.0914%)	-0.0257 (-0.448%)	-0.0286 (-0.499%)
KR	2.99	+0.0314 (+1.05%)	+0.193 (+6.45%)	+0.289 (+9.67%)
GR	4.712	+0.0731 (+1.55%)	+0.447 (+9.49%)	+0.655 (+13.9%)

Table 9

Changes in mean CV during sign-preserving interpolation from signed unit magnitudes to empirical *C. elegans* magnitudes. Baseline column lists the value at $f = 0$; other entries show absolute change followed by percent change.

Metric	Baseline	$f = 0.25$	$f = 0.5$	$f = 0.75$	$f = 1$
MC	0.2839	+0.00492 (+1.73%)	+0.0134 (+4.72%)	+0.0173 (+6.1%)	+0.021 (+7.39%)
IPC	0.2117	-0.00641 (-3.03%)	-0.00466 (-2.2%)	-0.00324 (-1.53%)	-0.00134 (-0.635%)
KR	0.2609	-0.00808 (-3.1%)	-0.00403 (-1.54%)	-0.000338 (-0.129%)	+0.00242 (+0.929%)
GR	0.4391	-0.00993 (-2.26%)	-0.00448 (-1.02%)	+0.000223 (+0.0509%)	+0.0034 (+0.775%)

Table 10

Changes in mean metric value during sign-preserving interpolation from signed unit magnitudes to empirical *C. elegans* magnitudes. Baseline column lists the value at $f = 0$; other entries show absolute change followed by percent change.

Metric	Baseline	$f = 0.25$	$f = 0.5$	$f = 0.75$	$f = 1$
MC	4.048	+0.0868 (+2.14%)	+0.193 (+4.76%)	+0.264 (+6.51%)	+0.304 (+7.5%)
IPC	6.121	+0.048 (+0.784%)	+0.154 (+2.51%)	+0.226 (+3.69%)	+0.27 (+4.4%)
KR	2.895	-0.0557 (-1.92%)	+0.0046 (+0.159%)	+0.0564 (+1.95%)	+0.0957 (+3.31%)
GR	4.558	-0.125 (-2.75%)	-0.0156 (-0.342%)	+0.0825 (+1.81%)	+0.156 (+3.42%)

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Table 11

Changes in mean CV during sign-preserving interpolation from signed unit magnitudes to shuffled empirical *C. elegans* magnitudes. Baseline column lists the value at $f = 0$; other entries show absolute change followed by percent change.

Metric	Baseline	$f = 0.25$	$f = 0.5$	$f = 0.75$	$f = 1$
MC	0.2839	+0.0143 (+5.05%)	+0.028 (+9.88%)	+0.0409 (+14.4%)	+0.0497 (+17.5%)
IPC	0.2117	+0.00991 (+4.68%)	+0.0157 (+7.44%)	+0.0247 (+11.7%)	+0.0303 (+14.3%)
KR	0.2609	+0.0226 (+8.68%)	+0.0491 (+18.8%)	+0.0689 (+26.4%)	+0.0836 (+32.1%)
GR	0.4391	+0.0317 (+7.23%)	+0.0662 (+15.1%)	+0.0902 (+20.5%)	+0.107 (+24.4%)

Table 12

Changes in mean metric value during sign-preserving interpolation from signed unit magnitudes to shuffled empirical *C. elegans* magnitudes. Baseline column lists the value at $f = 0$; other entries show absolute change followed by percent change.

Metric	Baseline	$f = 0.25$	$f = 0.5$	$f = 0.75$	$f = 1$
MC	4.048	+0.133 (+3.29%)	+0.319 (+7.87%)	+0.463 (+11.4%)	+0.583 (+14.4%)
IPC	6.121	+0.176 (+2.88%)	+0.434 (+7.09%)	+0.62 (+10.1%)	+0.783 (+12.8%)
KR	2.895	+0.257 (+8.88%)	+0.586 (+20.2%)	+0.861 (+29.7%)	+1.08 (+37.3%)
GR	4.558	+0.529 (+11.6%)	+1.22 (+26.8%)	+1.81 (+39.6%)	+2.28 (+49.9%)

Table 13

Changes in mean CV during sign-preserving interpolation from empirical *C. elegans* magnitudes to shuffled empirical magnitudes. Baseline column lists the value at $f = 0$; other entries show absolute change followed by percent change.

Metric	Baseline	$f = 0.25$	$f = 0.5$	$f = 0.75$	$f = 1$
MC	0.3049	-0.00427 (-1.4%)	-0.00478 (-1.57%)	+0.00836 (+2.74%)	+0.0287 (+9.43%)
IPC	0.2104	+0.00135 (+0.642%)	+0.00394 (+1.87%)	+0.0155 (+7.35%)	+0.0317 (+15%)
KR	0.2634	-0.00128 (-0.486%)	+0.00995 (+3.78%)	+0.0375 (+14.2%)	+0.0812 (+30.8%)
GR	0.4425	-0.000606 (-0.137%)	+0.0151 (+3.42%)	+0.0515 (+11.6%)	+0.104 (+23.4%)

Table 14

Changes in mean metric value during sign-preserving interpolation from empirical *C. elegans* magnitudes to shuffled empirical magnitudes. Baseline column lists the value at $f = 0$; other entries show absolute change followed by percent change.

Metric	Baseline	$f = 0.25$	$f = 0.5$	$f = 0.75$	$f = 1$
MC	4.352	-0.0815 (-1.87%)	-0.0999 (-2.3%)	-0.00326 (-0.075%)	+0.28 (+6.43%)
IPC	6.39	-0.0742 (-1.16%)	-0.0477 (-0.747%)	+0.121 (+1.89%)	+0.513 (+8.03%)
KR	2.991	-0.0327 (-1.09%)	+0.0777 (+2.6%)	+0.396 (+13.2%)	+0.985 (+32.9%)
GR	4.714	-0.0561 (-1.19%)	+0.181 (+3.83%)	+0.855 (+18.1%)	+2.12 (+45%)

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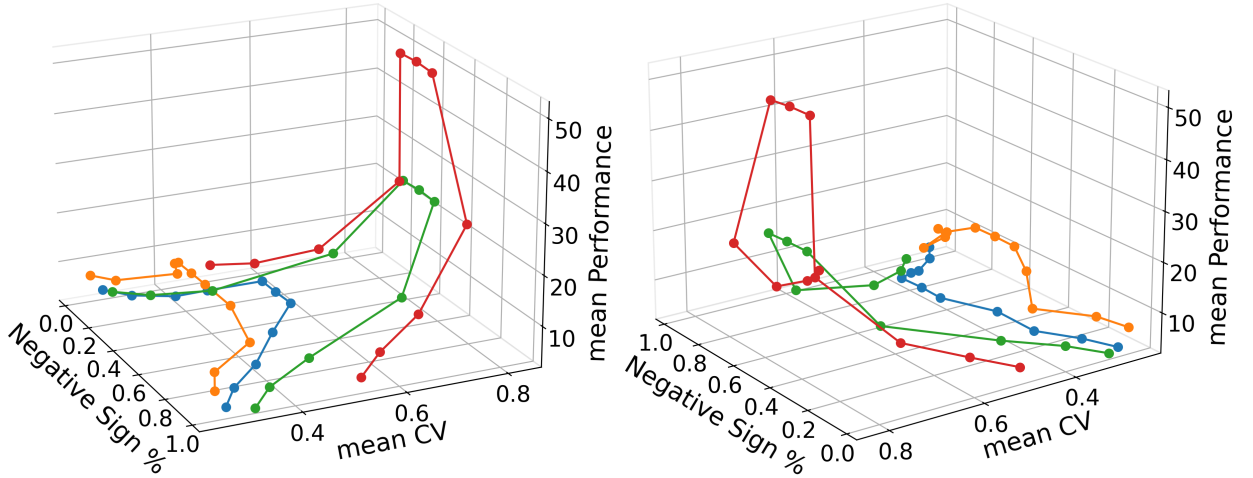


Figure 6: Mean performance and mean hyperparameter CV as a function of negative sign percentage for the *C. elegans*-matched ER topology. IPC is denoted in orange, GR in red, MC in blue, and KR in green. The same performance–invariance tradeoff appears without the biological topology, indicating that global sign balance can act as a primary control parameter within this matched random-topology setting. A total of 50 trials were performed for each fraction.

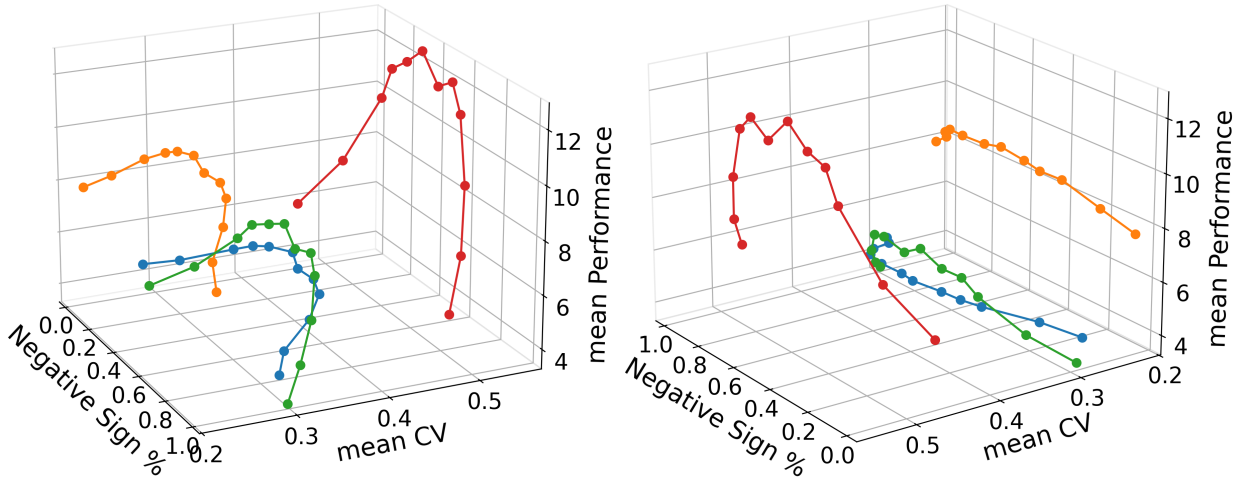


Figure 7: Mean performance and mean hyperparameter CV as a function of negative sign percentage for the empirical *C. elegans* magnitude pattern after removing connections with unknown or complex signs. IPC is denoted in orange, GR in red, MC in blue, and KR in green. Reintroducing inhibitory signs again traces the performance–invariance tradeoff, now with empirical magnitudes held in place. A total of 50 trials were performed for each fraction.

Sign-sweep quick look: the *C. elegans* sweep can cross below the original raw spectral radius before the empirical sign fraction

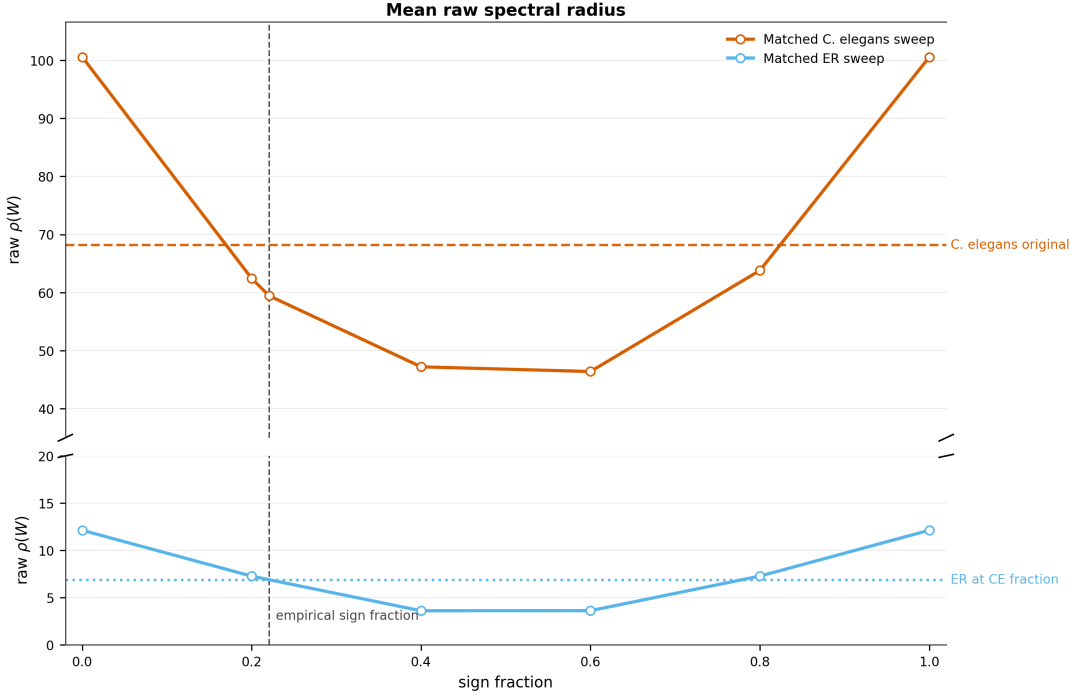


Figure 8: Quick look at the sign-sweep comparison between the *C. elegans* and *C. elegans*-matched ER topologies. The vertical dashed line marks the empirical negative-edge fraction. In the *C. elegans* sweep, the raw spectral radius can dip below the original baseline slightly before that fraction is reached because sign placement is simultaneously being randomized. The ER-matched sweep shows a smaller offset, consistent with signs already being effectively random relative to topology.

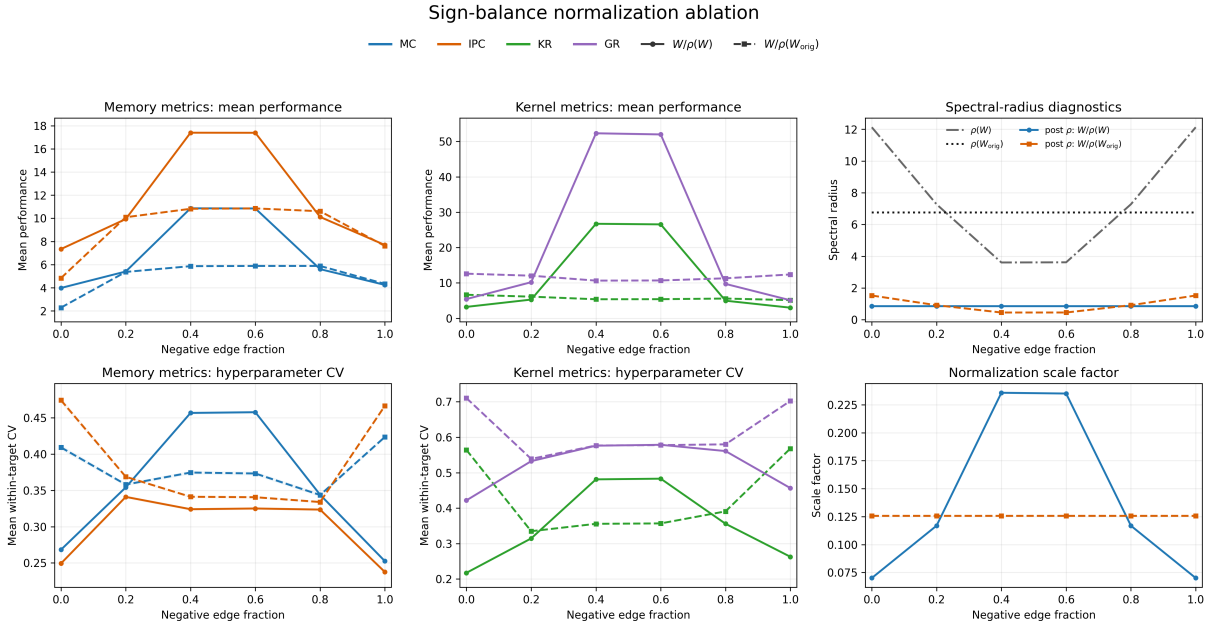


Figure 9: Sign-balance normalization ablation for the *C. elegans*-matched ER topology. Scaling by the original spectral radius weakens the performance increase associated with changing the negative edge fraction.

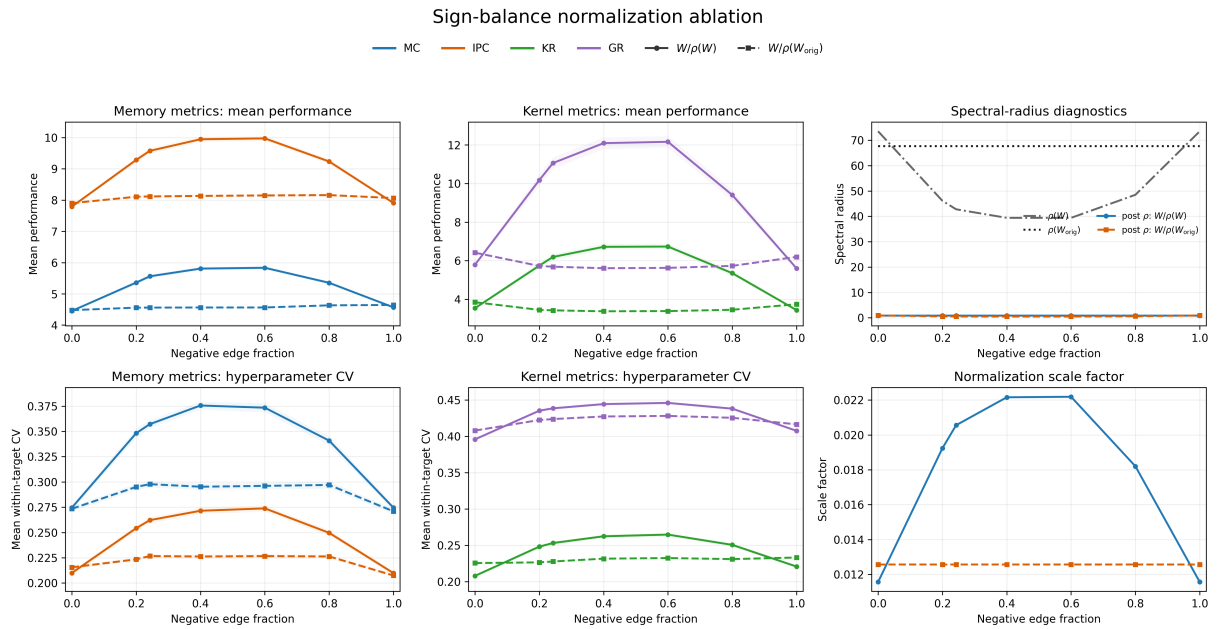


Figure 10: Sign-balance normalization ablation for the unknown/complex-removed *C. elegans* magnitude pattern after removing the original signs. Scaling by the original spectral radius weakens the performance increase associated with changing the negative edge fraction.

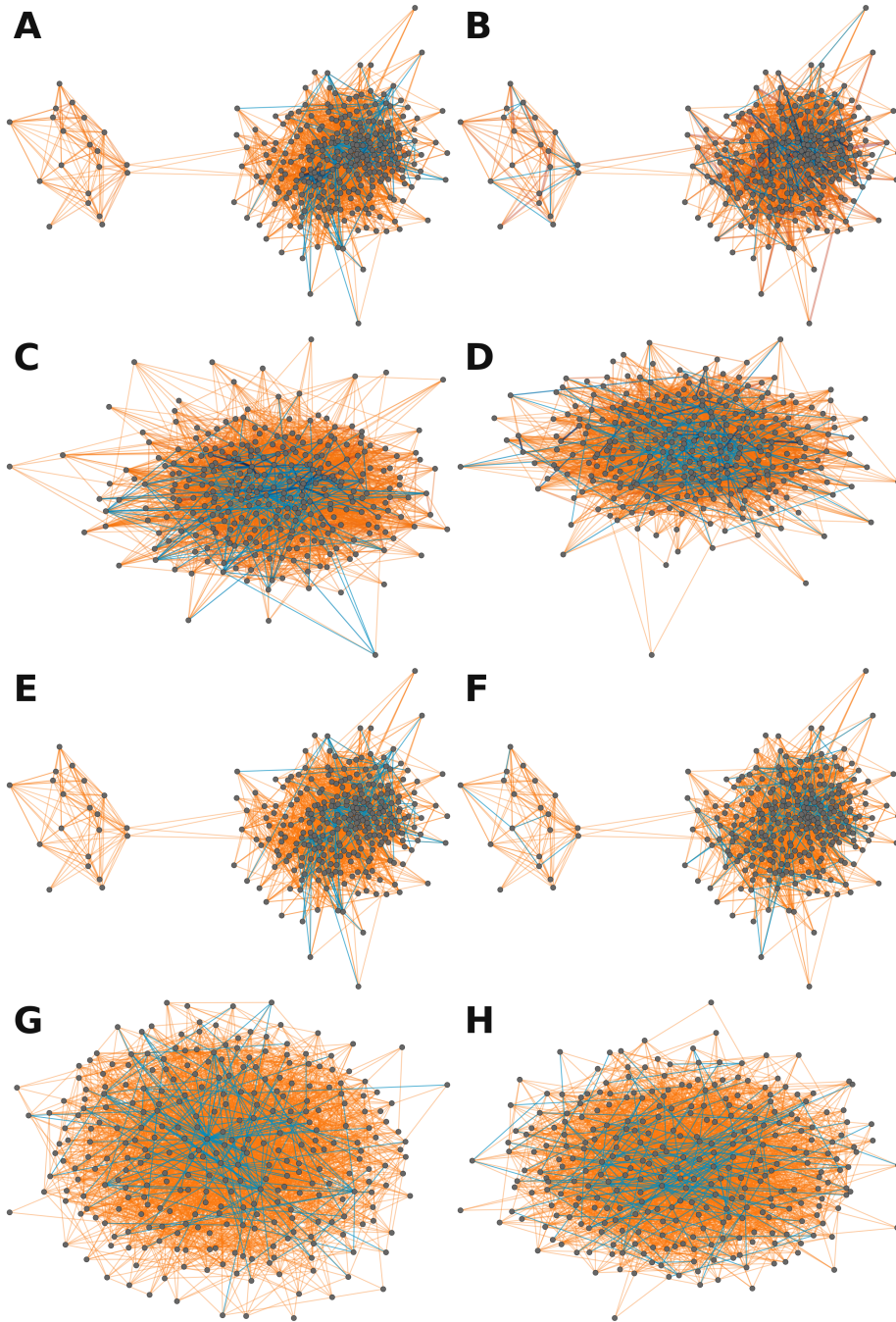


Figure 11: Network visualizations for architectures A–H. In all panels, nodes represent neurons, orange edges are excitatory synapses, blue edges are inhibitory synapses, and darker edges indicate stronger synaptic weights. A: original *C. elegans* connectome. B: original connectome with shuffled weights. C: connection-shuffled *C. elegans* connectome. D: connection-shuffled connectome with an additional weight shuffle. E: original topology with signed binary (± 1) weights determined by the original edge signs. F: panel E after sign shuffling. G: connection-shuffled version of panel E. H: panel G after sign shuffling.

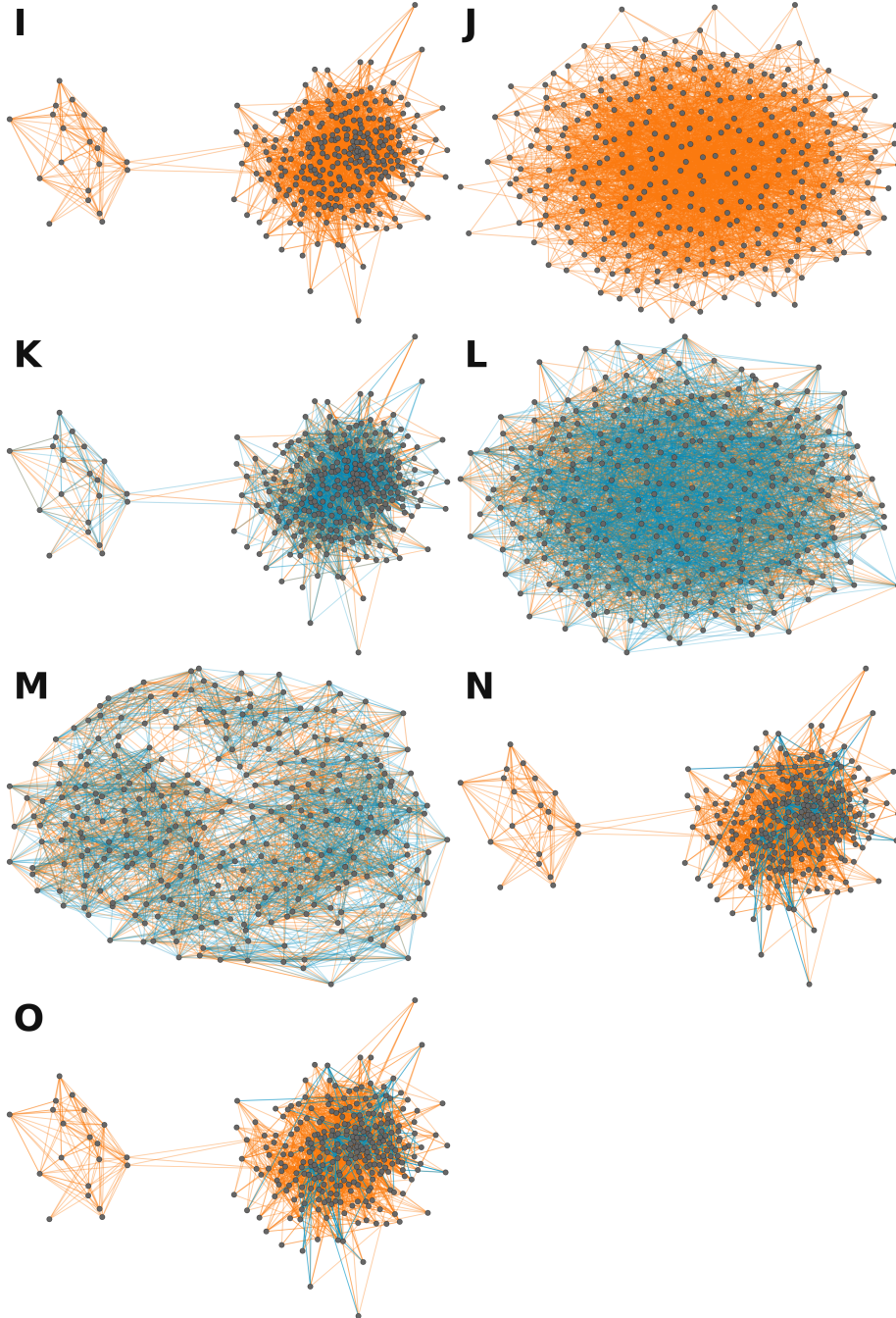


Figure 12: Additional control architectures I–O. I: *C. elegans* connectome with binary weights. J: binary-weight *C. elegans* connectome after connection shuffling. K: *C. elegans* topology with Gaussian weights. L: *C. elegans*-matched Erdős–Rényi network with Gaussian weights. M: small-world network with $p = 0.1$ and Gaussian weights. N: sign-preserving Gaussian model. O: sign-preserving uniform model.

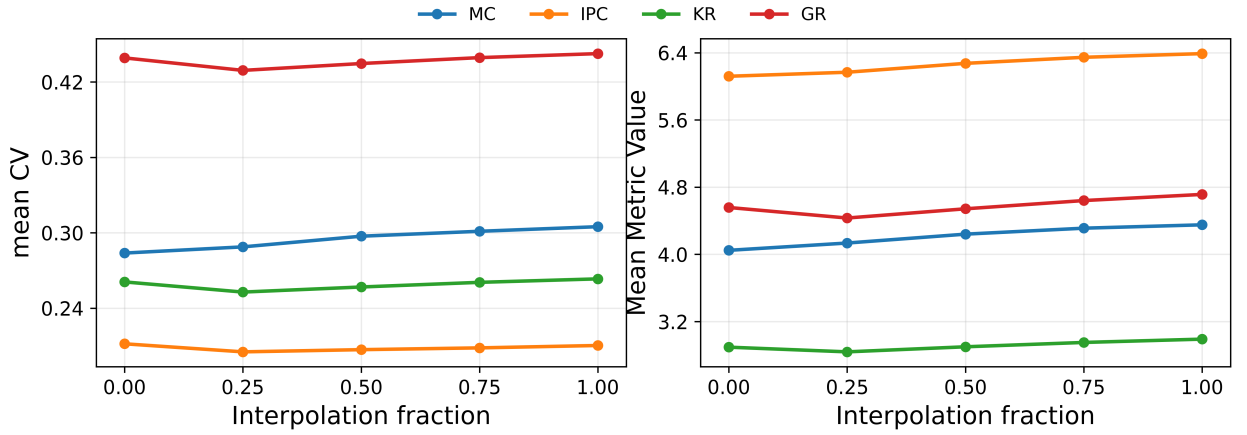


Figure 13: Sign-preserving interpolation from signed unit magnitudes to empirical *C. elegans* magnitudes. The original topology and edge signs are fixed for every interpolation fraction; only nonzero magnitudes change. Left: mean CV across the hyperparameter sweep. Right: mean task-agnostic metric value.

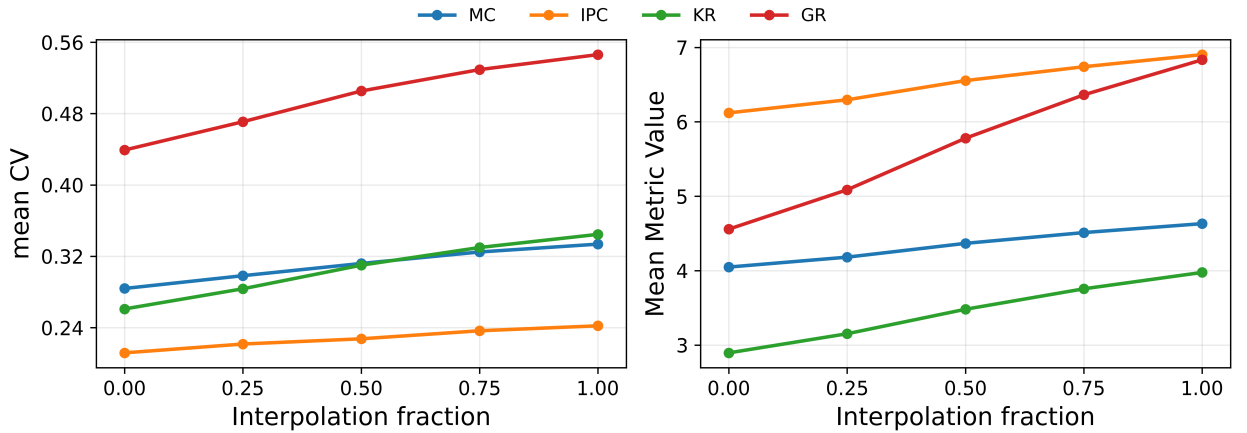


Figure 14: Sign-preserving interpolation from signed unit magnitudes to shuffled empirical *C. elegans* magnitudes. The endpoint uses a random permutation of the empirical magnitudes across the original edge set while preserving each edge's original sign. Left: mean CV across the hyperparameter sweep. Right: mean task-agnostic metric value.

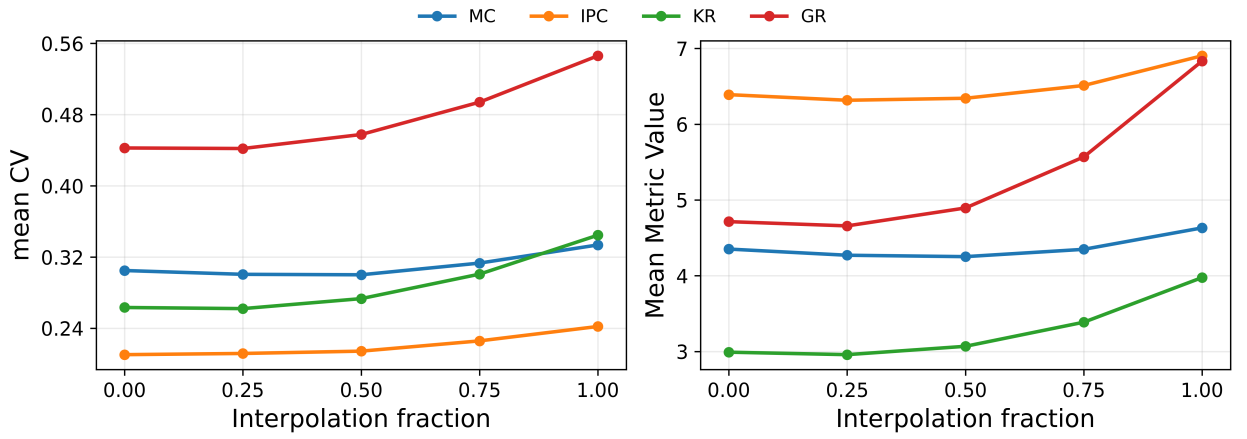


Figure 15: Sign-preserving interpolation from empirical *C. elegans* magnitudes to shuffled empirical magnitudes. The endpoint redistributes empirical magnitudes across the fixed topology, but every edge retains its original sign throughout the interpolation. Left: mean CV across the hyperparameter sweep. Right: mean task-agnostic metric value.